Equitable Routing and Scheduling for Mobile Vaccination Services Targeting Vulnerable Populations

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Abstract

During a pandemic, delivering vaccination campaigns that are both timely and inclusive is essential to prevent sharp infection surges. Yet, certain population groups, such as rural residents, people with mobility limitations (e.g., the elderly or disabled), and refugees, often struggle to access fixed vaccination sites, leading to lower uptake. To address this gap, fixed-site services can be complemented by mobile facilities (MFs) that bring vaccines directly to underserved communities. This study optimizes MF routes and schedules over a planning horizon to minimize the number of individuals unable to access vaccination due to capacity restrictions or delays, as well as the total traveling distance of MFs, while ensuring equitable service across locations. We develop a mathematical model and incorporate alternative equity approaches into the model. A case study on COVID-19 vaccination in Gaziantep, Türkiye, considers both Syrian refugees and Turkish citizens across diverse neighborhoods. Our findings show that carefully designed MF deployment strategies improve coverage among vulnerable groups substantially while keeping operational costs manageable. A comparison of different equity model solutions in the case study indicates that the Rawlsian approach provides the best trade-offs. The results provide actionable insights for public health authorities seeking to design equitable vaccination strategies under resource and time constraints.

Keywords: Mobile vaccination services, Equity-aware optimization, Multi-period routing and scheduling, Public health planning, Vulnerable populations, Scenario-based optimization

1. Introduction

Equitable access to vaccination is a longstanding public health challenge, particularly for individuals who face barriers such as limited mobility, socio-economic disadvantage, or refugee status. During the COVID-19 pandemic, many countries reported substantial vaccination disparities, with vulnerable groups experiencing lower uptake and delayed access [1]. Physical access constraints, information gaps, and administrative barriers further hindered immunization among disadvantaged populations [2]. These findings underscore the need for delivery models that actively reduce access gaps and support more inclusive health outcomes. Mobile vaccination units are a key means of reaching under-

served communities, and studies show that they significantly improve vaccine uptake when deployed strategically [3, 4, 5].

Türkiye provides a particularly relevant context: the country hosts more than 3 million Syrian refugees [6] and exhibits pronounced disparities in vaccination access. Survey evidence from the IFRC and Turkish Red Crescent indicates that refugees often encounter language barriers, difficulty securing appointments, and limited physical access to vaccination sites [7]. These challenges align with global patterns of under-immunization among refugee populations, driven by restricted healthcare access, overcrowded living conditions, and cultural or informational barriers [8, 9, 10]. Developing vaccination strategies that both improve coverage and effectively prioritize vulnerable groups is therefore an urgent policy priority.

To address this need, we develop a multi-period framework for planning mobile vaccination services that explicitly incorporates equity and uncertainty. The model represents population groups based on location, vulnerability, ideal vaccination periods, and preferences for vaccine types. We integrate and compare three alternative equity mechanisms—the *Gini Index*, the *Minimum Utility Threshold*, and the *D-Index*—to reflect differing perspectives on fairness in public policy and health equity [11, 12]. This framework enables planners to explore trade-offs between efficiency and equity and to design vaccination strategies that better serve underserved communities.

We formalize this setting as the Mobile Facility Routing and Scheduling Problem (MFRSP), linking tactical vaccine allocation with daily routing and operational scheduling under uncertain vaccine uptake. To ensure scalability and practical applicability, we propose a three-phase solution methodology. Phase I allocates vaccination capacity across population groups and time periods; Phase II constructs feasible daily routes for mobile teams; and Phase III generates scenario-based vaccination schedules that capture uncertainty in individuals' willingness or ability to be vaccinated. This structure supports real-world implementation across hundreds of neighborhoods and thousands of demographic groups while remaining interpretable for decision-makers.

We apply the framework to the COVID-19 vaccination campaign in Gaziantep, one of Türkiye's largest refugee-hosting provinces. Results from both illustrative and large-scale instances reveal several policy-relevant insights: modest expansions in mobile fleet capacity can substantially improve population coverage, and equity-focused strategies can meaningfully prioritize vulnerable groups with limited efficiency loss. These findings offer practical guidance for provincial health authorities seeking to design vaccination services that are both efficient and socially equitable.

The contributions of this study are fourfold. (1) **Equity-focused planning:** We propose a multi-period mobile vaccination planning model that embeds explicit equity metrics into allocation and scheduling decisions. (2) **Integration of uncertainty:** We incorporate scenario-based uptake uncertainty to reflect behavioral and informational barriers documented in the public health literature. (3) **Comparative equity mechanisms:** We incorporate horizontal and vertical equity using multiple alternative approaches and analyze their implications for service coverage, lateness, and operational burden. (4) **Policy-relevant insights:** Using detailed data from Gaziantep, we provide evidence on how mobile fleet size, prioritization rules, and equity goals influence vaccination outcomes for

vulnerable populations.

To systematically interpret the findings, our study adopts the Levesque Framework [13] as an *a posteriori* multidimensional lens on access to healthcare. While most operations research studies emphasize the organization and efficiency of supply-side decisions, the Levesque Framework highlights how health system design interacts with individuals' ability to access and utilize services through the dimensions such as Approachability, Acceptability, Availability and Accommodation, Affordability and Appropriateness [13]. By embedding equity considerations directly into the planning of mobile vaccination services, our approach proactively addresses several supply-side barriers faced by marginalized populations, thereby facilitating the translation of optimization-based decisions into actionable public health improvements. This perspective strengthens the policy relevance of the proposed framework by explicitly linking operational planning choices to broader access and equity objectives in public health. Overall, this research provides an analytical framework for designing equitable mobile vaccination strategies that can improve timely access to essential health services in socioeconomically disadvantaged communities during public health emergencies.

The remainder of the paper is organized as follows. Section 2 reviews related literature, Section 3 presents the problem definition, Section 4 introduces the solution methodology, Section 5 reports computational results, and Section 6 discusses the Gaziantep case study and associated managerial insights. Section 7 concludes the paper.

2. Literature Review

Operations Research methods have played a central role in supporting pandemic response through demand forecasting, vaccine supply chain design, and resource allocation [14, 15]. Most vaccination planning studies focus on allocation, inventory management, or siting of fixed vaccination centers [16, 17]. In contrast, this study examines the planning of mobile vaccination services aimed at reaching vulnerable populations—an area that has received comparatively less attention. Several studies optimize routes of mobile vaccination units but do not explicitly address equity [18, 19, 20]. In this section, we focus on models that incorporate equity considerations into the routing and scheduling of mobile vaccination services.

Equity remains a foundational yet multifaceted concept in public health planning. As Braveman [21] note, the term lacks a universal consensus. In health systems, equity generally refers to allocating resources proportionally to individuals' needs and vulnerabilities to achieve comparable health outcomes, whereas fairness emphasizes avoiding systematic bias in the delivery process [10]. In OR, equity has been modeled in diverse forms—from Rawlsian max–min principles to inequality index–based measures, such as the Gini coefficient [22]. Incorporating equity requirements into optimization models inevitably introduces efficiency–equity trade-offs [23], and perfect fairness is rarely attainable in practice [24].

A growing body of work embeds equity directly into vaccination models. Yet, as Şimşek et al. [10] argue, the specific needs of disproportionately underserved groups—such as migrants and refugees—remain insufficiently addressed. Equity in vaccination planning can be interpreted along two complementary

dimensions. *Horizontal equity* ensures that individuals or communities with comparable needs receive comparable access. Examples include enforcing Gini-based penalties and pro rata allocation rules within epidemiological frameworks [25], minimizing deviations from target equitable coverage levels [26], ensuring equitable travel times [27], and using equity-aware epidemiological models [28]. In contrast, *vertical equity* recognizes that disadvantaged or high-risk groups may require preferential treatment to achieve comparable health outcomes; thus, resource allocation may justifiably differ across groups in proportion to their level of need, such as prioritizing high-risk clusters for access to nearby facilities [29]. Effective vaccination strategies must therefore balance these principles by providing uniform service standards where appropriate, while directing additional support toward populations that face structural barriers such as limited mobility, poverty, or refugee status. Our study aims to achieve both horizontal equity across locations and vertical equity by prioritizing vulnerable populations.

Research on mobile health facilities can be broadly categorized into two streams. The first stream involves *mobile clinics*, where individuals travel to temporary service points. These studies emphasize strategic location decisions rather than routing. Examples include dynamic equitable placement of mobile suppliers based on socioeconomic indicators [30] or multi-period facility siting for en-route refugees with service continuity constraints [31, 32]. A second stream focuses on routing *mobile vaccination vehicles*. This literature includes models that incorporate booster doses [33], age-weighted unmet demand under epidemic dynamics [34], or maximize service provision for migrant farm workers [35]. While these studies address vulnerable populations, equity is typically addressed through weighting schemes rather than a systematic comparison of alternative equity mechanisms. Studies that do incorporate explicit prioritization—e.g., for elderly or disabled individuals [36]—often do so in single-day, simplified settings. Our study differs from Yucesoy et al. [36] in both problem characteristics—multi-period scheduling aspect for groups of patients, capacity restrictions, ideal vaccination timing, uncertainty—and the methodology presented.

A related line of research focuses on drone-based vaccine distribution, introducing robust or epidemic-aware equity constraints at the regional level [37, 38]. These models address access barriers in remote regions but do not capture human-centered, multi-period routing problems required for ground-based mobile clinics.

Overall, the literature lacks a unified framework that jointly integrates multi-period routing, operational scheduling, explicit equity mechanisms, ideal vaccination timing, and behavioral uncertainty, particularly for refugee and mobility-constrained populations. This study addresses this gap. Table 1 provides a comparative overview of existing studies with respect to multi-period planning, routing decisions, equity considerations, and real-world case applications.

Building on the gaps, we define a multi-period scheduling and routing problem that incorporates ideal vaccination periods, vaccine-type preferences, and the prioritization of disadvantaged groups. Our framework reflects how public health agencies allocate limited mobile resources across neighborhoods over time. It embeds eligibility policies through ideal vaccination periods pre-determined for different risk groups and models behavioral uncertainty using scenario-based show-up rates driven by hesitancy and information gaps. We evaluate three equity mechanisms (the Gini Index, a Rawlsian minimum

Table 1: Summary of studies on mobile vaccination planning

Study	Multi-period	Multi-dose	Capacity Consideration	Decision per Period	Uncertainty	Application Context	Equity Mechanism	Vulnerable Population
Chen et al. [30]	✓	×	×	Relocation	×	Deployment of mobile vaccine distribution centers	Min–max effective distance (reducing worst-off outcome)	Individuals with mobility disadvantage
Bayraktar et al. [31]	✓	×	×	Relocation	×	Mobile relief provision to refugees on the move	Continuity-of-service hard constraint	Refugees in transit
Pashapour et al. [32]	✓	×	✓ (MF capacity)	Relocation	×	Mobile relief aid with capacitated MFs	Continuity-of-service hard constraint	Refugees in transit
Santa González et al. [33]	✓	✓	✓ (MF capacity & service time)	Routing	×	Mobile health service routing	Weighted equity in objective	Disadvantaged rural villages
Vahdani et al. [34]	✓	✓	✓ (storage & vehicle capacity)	Routing	✓ (Demand)	Vaccine distribution to prioritized groups	Equity-adjusted penalty for unmet demand	Age-prioritized population groups
Salman et al. [35]	✓	×	×	Relocation	×	Mobile health services to migrant workers	Continuity + proximity constraints	Migrant agricultural workers
Wang et al. [37]	✓	✓	✓ (drone payload & range)	Routing	✓ (Supply variability)	Drone-based vaccine distribution	Area coverage parity constraint	Geographic regions
Sayarshad [38]	✓	×	✓ (drone payload & range)	Routing	×	Drone-based vaccine distribution	Risk-weighted equity	High-risk communities
Yucesoy et al. [36]	×	×	✓ (daily stock & service time)	Routing	×	Mobile vaccination routing in urban areas	Priority-weighted objective & distance thresholds	Older adults, disabled people
This Paper	✓	✓	✓ (daily MF capacity & service time)	Routing & scheduling (three-phase approach)	✓ (Demand scenarios)	Mobile vaccination service for vulnerable communities (Gaziantep Case)	Comparative analysis of multiple mechanisms: Gini Index, Minimum Utility Threshold, D-Index	Refugee and low-access population groups

utility threshold, and the D-index) to observe which approach provides more equitable solutions in our context.

While a growing number of vaccination distribution models address equity through optimization-based criteria, a significant research gap remains at the intersection of healthcare accessibility theory and operations research. A recent scoping review by [10] highlights that vulnerable groups often exhibit willingness to vaccinate, yet persistent barriers—such as information gaps, communication challenges, and system-level access constraints—continue to limit effective uptake in the absence of coordinated allocation programs. In the public health literature, the Levesque Framework of Access to Health Care is widely used as a qualitative, *a posteriori* tool to categorize such barriers across dimensions including Approachability and Acceptability [39, 10]. Consistent with this literature, our study adopts the Levesque Framework as an interpretive lens to contextualize the outcomes of a quantitative, multi-period optimization model, thereby linking operational planning decisions to broader access dimensions relevant for vulnerable populations.

To address the computational complexity of jointly optimizing all the decisions, we propose a scalable three-phase matheuristic that integrates tactical allocation, daily routing, and operational scheduling. A large-scale case study in Gaziantep, Türkiye, demonstrates that equity-aware deployment strategies can substantially improve vaccination coverage among vulnerable groups while maintaining operational efficiency.

This study provides the following core contributions: (1) incorporating public health–driven temporal prioritization through ideal vaccination periods; (2) embedding and comparing three equity mechanisms to quantify efficiency–equity trade-offs; (3) modeling behavioral uncertainty via scenario-based show-up rates; (4) developing a scalable three-phase solution framework; and (5) generating actionable insights on how equity criteria affect coverage, lateness, and resource utilization in real-world mobile vaccination deployments, interpreted through a healthcare accessibility lens.

3. Problem Definition

Motivated by the need to deliver vaccines efficiently while ensuring equitable access, we study the Mobile Facility Routing and Scheduling Problem (MFRSP). The problem is defined over a multi-period planning horizon, where mobile facilities (MFs) are routed to serve groups of individuals within each period. Mobile facility capacities are explicitly modeled, allowing patient groups to be partially served by multiple MFs, yielding a more realistic operational representation. We jointly minimize the total unvaccinated population, lateness, and travel distance through a normalized weighted objective reflecting public health priorities. Below, we introduce the main components of the problem.

Group Structure and Ideal Vaccination Periods. Vaccination demand is aggregated by groups based on their locations, ideal vaccination periods (IVP), vaccine type preferences, and vulnerability categories. The IVP indicates the time window during which individuals are eligible for vaccination, as determined by government priorities. Vulnerable groups (e.g., refugees, individuals with chronic diseases, or those in high-risk occupations for infection) are assumed to have been included in their ideal vaccination schedules through IVP by health authorities. In our case, we consider refugee groups as vulnerable

due to language barriers and fear of deportation, while Turkish citizen groups are considered non-vulnerable.

Let I represent the set of groups, partitioned into I^V (vulnerable groups) and I^{NV} (non-vulnerable groups), such that $I = I^V \cup I^{NV}$. Groups are assigned priority weights η^V and η^{NV}, with a higher value $\eta^V > \eta^{NV}$ to ensure vulnerable groups are prioritized. Each group $i \in I$ is associated with a location $l \in L$ via a binary parameter c_{il}, where the set L represents the locations of all groups. Each group has an IVP, drawn from the set T, which is determined by health officials and communicated to individuals based on vaccine allocation policies and vaccination history. We note that the unit of measure for the vaccination periods (VPs) may not be days. For example, in our case, the IVPs are in weeks due to Türkiye's COVID-19 vaccination procedures, while the time periods of the planning horizon correspond to days to accommodate the daily routing and vaccination schedule of MFs. The set of VPs is represented by T, whereas the set of days in the planning horizon is referred to as D. Because vaccination may occur later than the IVP, lateness is captured by parameter v_{id}, which measures the delay for group $i \in I$ vaccinated on day $d \in D$ of the planning horizon. Vaccination eligibility is tracked with parameter q_{id}, equal to 1 if day $d \in D$ is within or after the IVP of group $i \in I$. Maximum lateness thresholds γ^V and γ^{NV} are imposed for vulnerable and non-vulnerable groups, respectively. Lateness is measured by the number of VPs by which the vaccination for a group is late.

Planning Horizon. The planning horizon refers to the time interval consisting of multiple days during which vaccination services are scheduled and delivered. Since demand may extend beyond this boundary, two forms of spillover are considered: (i) unserved demand carried over from the previous horizon, and (ii) demand from the current horizon that can be deferred slightly into the next. To accommodate this flexibility, unserved demand from the previous horizon is incorporated into the current one, while the current planning horizon is extended by a limited number of additional VPs that serve as a buffer between consecutive horizons. For example, if the planning horizon spans 12 weeks, it may be extended to 13 weeks to accommodate a small fraction of current demand in the subsequent horizon. Without this extension, groups whose IVP falls in the 12th week would need to be served strictly during that week, leading to artificial demand accumulation toward the end of the horizon. Notably, no individual is assigned the 13th week as their IVP; this additional period serves solely as a scheduling buffer to smooth the transition to the end of the period. This mechanism prevents unrealistic clustering of demand at horizon boundaries and aligns with established practices in inventory management, where restrictive end-of-horizon assumptions are mitigated via continuity terms or rolling horizon planning. While spillovers are permitted to enhance scheduling flexibility, they are constrained by lateness limits that prevent excessive postponement and ensure vaccination schedules remain realistic and timely.

Vaccine Hesitancy Represented by Scenarios. In practice, vaccination uptake is uncertain due to hesitancy driven by cultural considerations, individual preferences, and misinformation. To address this uncertainty, we adopt a scenario-based modeling approach. Let S represent the set of scenarios, each with probability p_s. For group $i \in I$, n_{is} represents the number of individuals in group $i \in I$ who show up under scenario $s \in S$. In our case study, three scenarios are considered, reflecting higher,

lower, and baseline hesitancy levels, each with equal probability.

Operational Characteristics and Routing Structure of Mobile Facilities. Let M be the set of MFs providing vaccination services. Each facility $m \in M$ has a daily service capacity h_m (indicating an upper bound for its total travel and service time), maximum daily vaccination capacity v_m, starting and ending depot $j_m \in J$, where the set J represents the depot locations for all facilities, and feasible visit locations are $L_m = L \cup \{j_m\}$. Travel times between locations $j, l \in L_m$ are given by d_{jl}. Each service visit requires a setup time f before vaccinations begin and an average service time e per person. The maximum number of visits per day is limited by r, and δ denotes the unit travel cost.

At the start of each day, MF $m \in M$ departs from its depot j_m, visits a subset of locations $l \in L$ to vaccinate assigned groups, and returns to its depot the same day. Overnight stays at service locations are not allowed, ensuring daily replenishment of supplies (vaccines and medical equipment) and staff rotation.

Service Levels and Objectives. We define a service level (SL) as the prioritized proportion of individuals planned to be vaccinated in both vulnerable and non-vulnerable groups, represented by decision variables Q^V and Q^{NV}, respectively. The primary objective of the MFRSP is to maximize the overall SL across all groups. Additionally, there is a secondary objective of minimizing inequality in service delivery, enforced through one of the three approaches: bounding the *Gini Index*, applying a *minimum utility threshold*, and bounding the *D-Index*.

Figure 1 illustrates a sample solution for an MFRSP instance with two MFs originating from two different depots (represented by orange rectangles). Group locations are represented by yellow circles, with vulnerable groups indicated by smaller circles and non-vulnerable groups by smaller triangles within them. Colors indicate their IVPs (pinks have IVP on the first VP, i.e., week, and blues IVP on the second VP). Routes of the two MFS are shown by arcs: solid for day 1 and dashed for day 2, and blue for the first MF and red for the second MF. As Figure 1 demonstrates, in MFRSP, a group may be partially served by multiple MF visits across different days within the planning horizon.

To interpret the implications of our modeling choices beyond operational performance, we analyze results through the Levesque Framework of Access to Healthcare [13]. This framework provides a multidimensional lens—encompassing availability, accommodation, approachability, and appropriateness—through which the equity and accessibility impacts of mobile vaccination planning can be assessed. A detailed mapping between Levesque access dimensions and the quantitative indicators used in this study is provided in Table A.1 in Appendix A.

4. Solution Methodology for the Mobile Facility Routing and Scheduling Problem (MFRSP)

The MFRSP is inherently complex because it requires the simultaneous coordination of daily routing decisions and detailed operational scheduling under uncertain vaccine uptake in a multi-period planning horizon, group-specific lateness allowances, vulnerability prioritization, and equity constraints. Our computational experiments indicate that even small instances of the fully integrated model become intractable when attempted as a single optimization problem. To make the problem solvable at

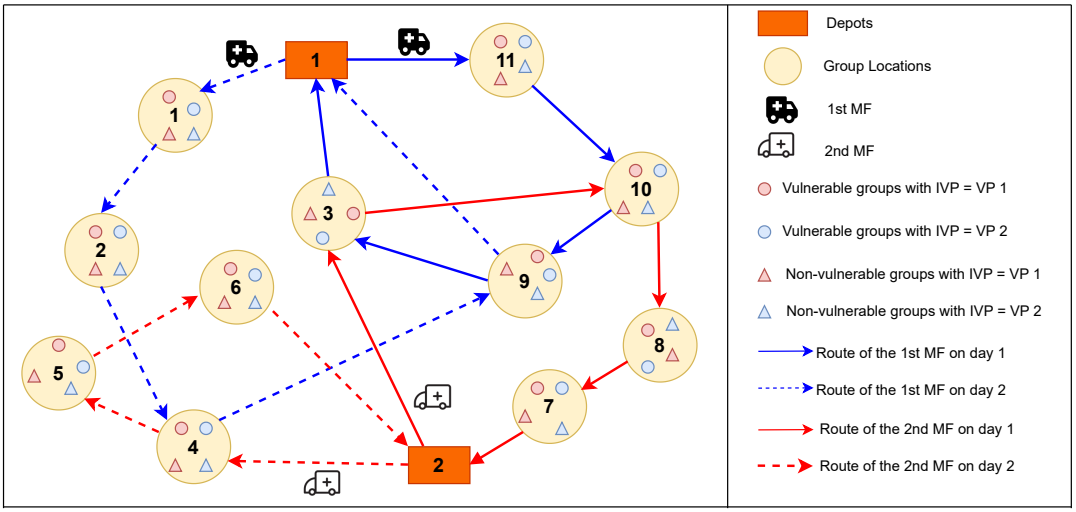


Figure 1: Sample MFRSP solution showing MF routes from two depots, group locations by vulnerability and IVP, and two-day routing (solid/dashed).

a real-world scale and produce outputs that are interpretable and implementable for health authorities, we propose a three-phase methodology that decomposes planning and operations into tractable subproblems, each formulated as a mathematical programming model.

Phase I: Tactical allocation across vaccination periods. Determines, for each group, how many individuals will be vaccinated in each vaccination period (VP), without assigning specific days or routes. This yields an aggregate allocation over the planning horizon.

Phase II: Routing per day. Given Phase I allocations by location and VP, constructs daily routes for mobile facilities (MFs) and determines day-level vaccination quantities for MFs while respecting travel/service capacities.

Phase III: Operational scheduling by MF and day under scenarios. Fixes routes from Phase II and produces the detailed day-by-day schedule, specifying how many individuals from each group each MF vaccinates on each day, under the scenario framework.

The following notation is used throughout the three phases.

Sets:

I^V	Vulnerable groups
I^{NV}	Non-vulnerable groups
I	All groups, $I = I^V \cup I^{NV}$
M	Mobile facilities
L	Group locations
J	Depots (with $j_m \in J$ the depot of MF m)
L_m	Locations MF $m \in M$ may visit, $L_m = L \cup \{j_m\}$
T	Vaccination periods (VPs)
D	Days in the planning horizon
D_t	Days in VP $t \in T$, where $D_t \cap D_{t'} = \emptyset$, for any $t, t' \in T$ and $D = \bigcup_{t \in T} D_t$
S	Scenarios

Parameters:

n_{is}	population of group $i \in I$ under scenario $s \in S$
q_{it}	1 if VP $t \in T$ is within or later than the IVP of group $i \in I$, 0 otherwise
$\tilde{q}_{id}$	1 if day $d \in D$ is within or later than the IVP of group $i \in I$, 0 otherwise
v_{it}	Lateness of vaccinating group $i \in I$ in VP $t \in T$
$\tilde{v}_{id}$	Lateness of vaccinating group $i \in I$ on day $d \in D$
c_{il}	1 if group $i \in I$ is located at $l \in L$, 0 otherwise
p_s	Probability of occurrence of scenario $s \in S$
f	Setup time required before MF provides service at a location
e	Time required to vaccinate one individual
r	Maximum number of locations an MF can visit in a day
α	Number of service days per VP per MF
h_m	Daily service capacity (time) of MF $m \in M$
a_m	Maximum number of vaccinations that MF $m \in M$ can administer in a day
j_m	Starting and ending depot of MF $m \in M$
δ	Unit cost of distance traveled by an MF
η^V	Priority level assigned to vulnerable groups
η^{NV}	Priority level assigned to non-vulnerable groups
γ^V	Maximum allowable lateness (proportion) for vulnerable groups
γ^{NV}	Maximum allowable lateness (proportion) for non-vulnerable groups
d_{jl}	Travel time between locations $j, l \in \bigcup_{m \in M} L_m$ (in minutes)
$\bar{d}$	Travel-time metric used in Phase I

Decision Variables:

W^I_{its}	Number of individuals in group $i \in I$ vaccinated during VP $t \in T$ under scenario $s \in S$ (Phase I)
W^{II}_{mids}	Number of individuals in group $i \in I$ vaccinated on day $d \in D$ by MF $m \in M$ under scenario $s \in S$ (Phases II& III)
A^I_{lt}	1 if location $l \in L$ is visited in VP $t \in T$, 0 otherwise (Phase I)
A^{II}_{mld}	1 if MF $m \in M$ visits location $l \in L$ on day $d \in D$, 0 otherwise (Phase II)
Q^V	Service level achieved for vulnerable groups (Phases I & III)
Q^{NV}	Service level achieved for non-vulnerable groups (Phases I & III)
Y_{mjld}	1 if MF $m \in M$ travels arc (j, l) on day d ($j, l \in L_m$), 0 otherwise (Phase II)
Z_{mjd}	Miller–Tucker–Zemlin (MTZ) visiting order variable for MF $m \in M$ at node $j \in L_m$ on day $d \in D$ (for subtour elimination in Phase II)

4.1. Phase I: Scheduling of vaccination of groups

Phase I determines W^I_{its} (the number of individuals from group i vaccinated in VP $t \in T$) and A^I_{lt} (the set of locations served in VP $t \in T$) so as to maximize the service level (SL). Routing decisions are not modeled in this phase. Instead, daily travel time is approximated by a proxy metric $\bar{d}$. In our computational experiments, we initially set $\bar{d}$ to the average of all pairwise travel times d_{jl} with $j, l \in \bigcup_{m \in M} L_m$ and $j \neq l$, and iteratively update it if required (see Section 4.5).

We utilize the following model in the first phase.

$$\max \quad \eta^V Q^V + \eta^{NV} Q^{NV} \tag{1}$$

subject to:

$$\frac{\sum_{i\in I^V}\sum_{t\in T}\sum_{s\in S} p_s W^I_{its}}{\sum_{i\in I^V}\sum_{s\in S} p_s n_{is}} = Q^V \tag{2}$$

$$\frac{\sum_{i\in I^{NV}}\sum_{t\in T}\sum_{s\in S} p_s W^I_{its}}{\sum_{i\in I^{NV}}\sum_{s\in S} p_s n_{is}} = Q^{NV} \tag{3}$$

$$\frac{\sum_{i\in I^V}\sum_{t\in T}\sum_{s\in S} p_s v_{it} W^I_{its}}{\sum_{i\in I^V}\sum_{s\in S} p_s v_{i,|T|} n_{is}} \leq \gamma^V \tag{4}$$

$$\frac{\sum_{i\in I^{NV}}\sum_{t\in T}\sum_{s\in S} p_s v_{it} W^I_{its}}{\sum_{i\in I^{NV}}\sum_{s\in S} p_s v_{i,|T|} n_{is}} \leq \gamma^{NV} \tag{5}$$

$$\bar{d}(\sum_{l\in L} A^I_{lt} + 1) + f\sum_{l\in L} A^I_{lt} + e\sum_{i\in I} p_s W^I_{its} \leq \alpha \sum_{m\in M} h_m \qquad \forall t \in T, \forall s \in S \tag{6}$$

$$W^I_{its} \leq c_{il} n_{is} A^I_{lt} \qquad \forall i \in I, t \in T, s \in S \tag{7}$$

$$\sum_{t\in T} W^I_{its} \leq n_{is} \qquad \forall i \in I, s \in S \tag{8}$$

$$W^I_{its} \leq n_{is} q_{it} \qquad \forall i \in I, t \in T, s \in S \tag{9}$$

$$W^I_{its} \geq 0 \qquad \forall i \in I, t \in T\, s \in S \tag{10}$$

$$Q^V, Q^{NV} \geq 0 \tag{11}$$

$$A^I_{lt} \in \{0, 1\} \qquad \forall l \in L, t \in T \tag{12}$$

$$\text{Equity constraints of selected approach} \tag{13}$$

The model maximizes the overall TSL for both vulnerable and non-vulnerable groups, as defined in Objective (1). Constraints (2) and (3) specify the TSLs as the proportion of individuals receiving vaccination services for both vulnerable and non-vulnerable groups, respectively. Constraints (4) and (5) impose vaccination lateness bounds for vulnerable and non-vulnerable groups, respectively. Constraints (6) approximate weekly VP capacity using $\bar{d}$ under each scenario. Constraints (7) link location-level vaccinations with group-level allocations, ensuring consistency between the number of people served at a location in a given week and the total served across groups located there. Constraints (8) guarantee that the cumulative number of vaccinations in each group does not exceed its population. Constraints (9) enforce temporal feasibility by allowing vaccinations only within or after each group's IVP. Constraints (10) - (12) define the domains of the variables. Finally, Constraint (13) integrates the equity considerations, which are detailed in the subsequent section.

4.2. Phase II: Routing of each MF

Given optimal A'_{lt} and W'_{its} from Phase I, Phase II builds day–level routes and disaggregates W'_{its} across days and MFs. For each VP $t \in T^P$, let L_t represent the set of locations that will be visited in week $t \in T$, i.e., $L_t = \{l \in L : A'_{lt} = 1\}$. Phase II then determines (i) the sequence of locations visited by each MF on each day, represented by the routing variables Y_{mjld}, and (ii) the number of

individuals in each group served by each MF on each day under each scenario, represented by W^{II}_{mids}. These decisions are made separately for each VP $t \in T$.

The routing model to be solved for each VP $t \in T$ is as follows:

$$\max \sum_{i\in I} \sum_{m\in M} \sum_{d\in D_t} \sum_{s\in S} p_s W^{II}_{mids} \tag{14}$$

subject to:

$$\sum_{l\in L_t} Y_{m,j_m,l,d} = 1 \qquad \forall m \in M, d \in D_t \tag{15}$$

$$\sum_{l\in L_t} Y_{m,l,j_m,d} = 1 \qquad \forall m \in M, d \in D_t \tag{16}$$

$$\sum_{j\in L_t\cup j_m} Y_{mjld} - \sum_{j\in L_t\cup j_m} Y_{mljd} = 0 \qquad \forall m \in M, l \in L_t, d \in D_t \tag{17}$$

$$\sum_{j\in L_t\cup j_m} Y_{mjld} = 0 \qquad \forall m \in M, l \notin L_t, d \in D_t \tag{18}$$

$$\sum_{j\in L_t\cup j_m} \sum_{m\in M} \sum_{d\in D_t} Y_{mjld} \geq 1 \qquad \forall l \in L_t \tag{19}$$

$$\sum_{j\in L_t} Y_{mjld} = A^{II}_{mld} \qquad \forall m \in M, l \in L_t, d \in D_t \tag{20}$$

$$\sum_{j\in L_t} \sum_{l\in L_t} d_{jl} Y_{mjld} + f \sum_{l\in L_t} A^{II}_{mld} + e \sum_{i\in I} W^{II}_{mids} \leq h_m \qquad \forall m \in M, d \in D_t, s \in S \tag{21}$$

$$\sum_{i\in I} W^{II}_{mids} \leq a_m \qquad \forall m \in M, d \in D_t, s \in S \tag{22}$$

$$W^{II}_{mids} \leq \sum_{j\in L_t} \sum_{l\in L_t} c_{il} W'_{its} Y_{mjld} \qquad \forall m \in M, i \in I, d \in D_t, s \in S \tag{23}$$

$$\sum_{m\in M} \sum_{d\in D_t} W^{II}_{mids} \leq W'_{its} \qquad \forall i \in I, s \in S \tag{24}$$

$$Z_{mjd} - Z_{mld} + (r+1) Y_{mjld} \leq r \qquad \forall m \in M, j, l \in L_t, d \in D_t \tag{25}$$

$$Z_{m,j_m,d} = 0 \qquad \forall m \in M, d \in D_t \tag{26}$$

$$A^{II}_{mld}, Y_{mjld} \in \{0,1\} \qquad \forall m \in M, j, l \in L_t, d \in D_t \tag{27}$$

$$W^{II}_{mids} \geq 0 \qquad \forall m \in M, i \in I, d \in D_t, s \in S \tag{28}$$

$$Z_{mjd} \geq 0, \text{ int} \qquad \forall m \in M, j \in J, d \in D_t \tag{29}$$

The model maximizes the expected number of people vaccinated in a VP $t \in T^P$ across all scenarios, as defined in the objective function (14). Constraints (15) require each MF to depart from its origin at the beginning of the day, while Constraints (16) enforce a return to the same origin by the end of the day. Constraints (17) guarantee flow conservation at all visited locations. Constraints (18) and (19) restrict MF routes to the subset of locations preselected in Phase I. We note that in large-scale instances, considering all potential locations simultaneously would render the routing problem computationally intractable. Therefore, Phase I serves as a tactical filtering stage that identifies the locations prioritized

for the current planning horizon and Constraints (18) ensure that Phase II focuses exclusively on those selected locations, thereby reducing computational complexity while preserving the consistency of decisions across phases. Constraint (19) uses an inequality rather than an equality because a location may need to be visited multiple times—on different days and possibly by different MFs—when the number of individuals to be vaccinated at that location exceeds a single vehicle's daily capacity. Constraints (20) determine the set of locations visited by each MF, while Constraints (21) impose the daily working time limit for the MF. Constraints (22) enforce vaccine capacity limits by ensuring that the number of administered doses per day does not exceed each MF's maximum daily capacity under any scenario. Constraints (23) and (24) ensure consistency with Phase I by preventing an MF from administering more vaccines than are allocated for that purpose. Constraints (25) and (26) establish the visiting order of locations for each MF using Miller–Tucker–Zemlin (MTZ) subtour elimination constraints [40]. Finally, Constraints (27)-(29) define the domains of the decision variables.

4.3. Phase III (Scenario-based daily vaccination scheduling)

Phase III fixes the routes Y'_{mjld} and visits A'_{mld} determined in Phase II, and further refines the allocation variables W^{II}_{mids} by distributing vaccinations across days, MFs, and scenarios. Equity considerations are explicitly incorporated at this stage, with the objective of maximizing the prioritized service levels for both vulnerable and non-vulnerable groups. Individuals assigned to vaccination periods $t \in T^{NP}$ in Phase I are deferred to the subsequent rolling horizon.

Accordingly, we employ the following model in the third phase.

$$\max \quad \eta^V Q^V + \eta^{NV} Q^{NV} \tag{30}$$

subject to:

$$\frac{\sum_{i\in I^V, d\in D, s\in S, m\in M} p_s\, W^{II}_{mids}}{\sum_{i\in I^V, s\in S} p_s\, n_{is}} = Q^V \tag{31}$$

$$\frac{\sum_{i\in I^{NV}, d\in D, s\in S, m\in M} p_s\, W^{II}_{mids}}{\sum_{i\in I^{NV}, s\in S} p_s\, n_{is}} = Q^{NV} \tag{32}$$

$$\frac{\sum_{i\in I^V} \sum_{t\in T} \sum_{s\in S} \sum_{m\in M} p_s \tilde{v}_{id} W^{II}_{mids}}{\sum_{i\in I^V} \sum_{s\in S} p_s \tilde{v}_{i,|D|} n_{is}} \le \gamma^V \tag{33}$$

$$\frac{\sum_{i\in I^{NV}} \sum_{t\in T} \sum_{s\in S} \sum_{m\in M} p_s \tilde{v}_{id} W^{II}_{mids}}{\sum_{i\in I^{NV}} \sum_{s\in S} p_s \tilde{v}_{i,|D|} n_{is}} \le \gamma^{NV} \tag{34}$$

$$\sum_{j\in L_m} \sum_{l\in L_m} (d_{jl} Y'_{mjld} + f A'_{mld}) + e \sum_{i\in I} W^{II}_{mids} \le h_m \qquad \forall m \in M, d \in D, s \in S \tag{35}$$

$$W^{II}_{mids} \le c_{il} n_{is} A'_{mld} \qquad \forall i \in I, d \in D, s \in S, m \in M, l \in L_m \tag{36}$$

$$\sum_{d\in D} \sum_{m\in M} W^{II}_{mids} \le n_{is} \qquad \forall i \in I, s \in S \tag{37}$$

$$W^{II}_{mids} \le n_{is} \tilde{q}_{id} \qquad \forall i \in I, d \in D, s \in S, m \in M \tag{38}$$

$$W^{II}_{mids} \ge 0 \qquad \forall i \in I, d \in D\, s \in S \tag{39}$$

$$Q^V, Q^{NV} \ge 0 \tag{40}$$

Equity constraints of selected approach (41)

The model maximizes the overall TSL achieved across both vulnerable and non-vulnerable groups as defined in Objective (30). Constraints (31) and (32) specify the TSLs as the proportion of individuals receiving vaccination services for both vulnerable and non-vulnerable groups, respectively. Constraints (33) and (34) limit the allowable vaccination lateness for vulnerable and non-vulnerable groups, respectively, ensuring that delays remain within acceptable bounds. Constraints (35) impose the daily working time limit for each MF, while constraints (36) ensure that individuals can receive vaccination services on a particular day only if an MF is scheduled to visit their location on that day. Constraints (37) guarantee that the total number of vaccinated individuals in each group over the entire planning horizon does not exceed the group's population. Constraints (38) maintain temporal feasibility by allowing vaccinations only within or after the group's IVP, preventing premature vaccinations. Finally, constraints (39) - (40) define the domains of the decision variables, while constraints (41) incorporate the equity considerations, which are elaborated in the following section.

4.4. Equity Consideration

This study aims to integrate efficiency and equity concerns in the delivery of vaccination services through MFs. Efficiency is addressed by the model's primary objective functions, which seek to maximize overall service coverage and minimize delays. However, equitable distribution of vaccination services may still pose challenges. For instance, while some groups may receive vaccination services during their IVP, others may experience delays or remain unserved within the current planning horizon.

Chen and Hooker [41] presented a novel approach to balancing fairness and efficiency in optimization models, offering a set of social welfare functions that combine Rawlsian Leximax fairness and utilitarianism. These functions are defined by a single parameter, enabling their practical implementation. The method is demonstrated on healthcare resource allocation and disaster preparation problems, yielding reasonable solution times. Chen and Hooker [42] reviewed various mathematical schemes for incorporating fairness considerations into optimization models, covering inequality measures. The authors discussed practical approaches for formulating these criteria within linear, nonlinear, or mixed integer programming models and considered philosophical and ethical literature. Dönmez et al. [43] reviewed the challenge of incorporating fairness into mathematical models for humanitarian logistics due to the subjectivity of equity preferences and problem dynamics, resulting in various approaches in humanitarian decision-making. The paper presented a comprehensive review of existing fairness-related studies in humanitarian logistics, offering a structured categorization and a generic modeling framework for allocation, routing, and inventory decisions. It highlights the significant impact of the choice of equity function on allocation outcomes and explores alternative approaches that have remained unexplored. From another perspective, Alem et al. [44] discussed the importance of modeling equity in humanitarian logistics and highlighted the widespread use of the Gini coefficient. They introduced an approach to optimize the trade-off between effectiveness and equity using the original Gini coefficient via the Lorenz curve and emphasizing the differences in decisions between the original Gini measure and alternative definitions.

We assume that individuals who are not vaccinated within the current planning horizon will certainly be vaccinated in the subsequent one. Additionally, these individuals are expected to be vaccinated, on average, midway through the next planning horizon. Based on this assumption, the lateness of individuals who remain unvaccinated is defined as 1.5 times the number of IVPs in the current planning horizon. This lateness is quantified by a new parameter, θ. For example, if there are 12 IVPs in the current planning horizon, θ equals 18.

To capture the equitable distribution of vaccination services, we define U_i as the utility of the vaccination service received by group i. The utility functions for Phases I and III are provided in Equations (42) and (43), respectively.

$$U_i = 1 - \frac{\sum_{t\in T}\sum_{s\in S} p_s v_{it} W^I_{its} + \theta \sum_{s\in S} p_s (n_{is} - \sum_{t\in T} W^I_{its})}{\theta \sum_{s\in S} p_s n_{is}} \qquad \forall i \in I \qquad (42)$$

$$U_i = 1 - \frac{\sum_{m\in M}\sum_{d\in D}\sum_{s\in S} p_s \tilde{v}_{id} W^{II}_{mids} + \theta \sum_{s\in S} p_s (n_{is} - \sum_{m\in M}\sum_{d\in D} W^{II}_{mids})}{\theta \sum_{s\in S} p_s n_{is}} \qquad \forall i \in I \qquad (43)$$

In these equations, the first term in the numerator accounts for vaccinated individuals who experience lateness, while the second term captures unvaccinated individuals, each weighted by θ, representing the lateness associated with remaining unserved. The numerator thus represents the total disutility of group i, normalized by the maximum possible disutility (if no member were vaccinated within the planning horizon). Subtracting this fraction from 1 yields the utility of group i. Consequently, $U_i = 1$ indicates full and timely vaccination, $U_i = 0$ indicates no vaccination, and intermediate values correspond to partial or delayed service.

For clarity, these equations can be reformulated as Equations (44) and (45), respectively, where $\theta - v_{it}$ (or $\theta - \tilde{v}_{id}$) represents the effective contribution to utility for each vaccinated individual:

$$U_i = \frac{\sum_{t\in T}\sum_{s\in S} (\theta - v_{it}) W^I_{its}}{\theta \sum_{s\in S} n_{is}} \qquad \forall i \in I \qquad (44)$$

$$U_i = \frac{\sum_{m\in M}\sum_{d\in D}\sum_{s\in S} (\theta - \tilde{v}_{id}) W^{II}_{mids}}{\theta \sum_{s\in S} n_{is}} \qquad \forall i \in I \qquad (45)$$

Equity among groups is then incorporated into the model through one of three approaches: (i) constraining the overall inequity level via a bound on the *Gini Index*, (ii) ensuring a *Minimum Utility Threshold* across all groups, or (iii) bounding inequity relative to the population-weighted average utility using the *D-Index*.

The Gini Index approach aims for **horizontal equity** by penalizing unequal service among groups with comparable vaccination needs. The Minimum Utility Threshold approach captures **vertical equity** by ensuring that the least-advantaged groups receive a minimum level of service, directing resources toward higher-need populations. The D-index blends both perspectives by measuring deviations from a need-adjusted ideal distribution, encouraging uniformity among similar groups (horizontal equity) while prioritizing groups with greater vulnerability (vertical equity).

The additional parameters and decision variables utilized across these approaches are defined below.

Parameters:

θ	Lateness assigned to individuals not vaccinated within the planning horizon
ψ	Upper bound on inequity among the groups for the Gini Index
u_{min}	Minimum acceptable utility level across all groups
D	Upper bound on inequity among the groups for the D-Index
ϵ_i	Proportion of group $i \in I$ within the total population

Decision Variables:

U_i	Utility of vaccination service received by group $i \in I$
U_{avg}	Average utility across all groups
$B_{ii'}$	Absolute difference between utilities of groups i and i', $i, i' \in I$
F_i	Absolute deviation of group $i \in I$'s utility from the average utility, U_{avg}

4.4.1. Equity Consideration with a constraint that bounds the Gini Index

The Gini index, originally developed to measure income and wealth inequality, has been widely adopted in humanitarian logistics to quantify fairness in the allocation of scarce resources (Dastgoshade et al. [45], Alem et al. [44], Enayati and Özaltın [28], Gutjahr and Fischer [46]). In this study, the measure is used to assess inequality in the distribution of vaccination services among different groups. The corresponding social welfare function, denoted as $W(U)$, is given by equation (46).

$$W(U) = \frac{1}{2U_{avg}|I|^2} \sum_{i,i' \in I} |U_i - U_{i'}| \tag{46}$$

where U_{avg} represents the average utility across all groups and is defined as

$$U_{avg} = \frac{\sum_{i \in I} U_i}{|I|} \tag{47}$$

We introduce a parameter ψ to specify an upper bound on the acceptable level of inequity among the groups. To ensure that inequity remains below this threshold, the following constraint, (48), is added to the mathematical model of MFRSP:

$$\frac{1}{2U_{avg}|I|^2} \sum_{i \in I} \sum_{i' \in I} |U_i - U_{i'}| \leq \psi \tag{48}$$

To preserve linearity, Constraint (48) is reformulated using an auxiliary variable $B_{ii'}$, which represents the absolute difference between the utilities of groups i and i'. The nonlinear term $|U_i - U_{i'}|$ is thus replaced by a set of linear constraints (49)-(51):

$$\sum_{i \in I} \sum_{i' \in I} B_{ii'} \leq 2\ \psi\ U_{avg}\ |I|^2 \tag{49}$$

$$U_i - U_{i'} \leq B_{ii'} \qquad \forall i, i' \in I \tag{50}$$

$$U_{i'} - U_i \leq B_{ii'} \qquad \forall i, i' \in I \tag{51}$$

In this approach, the modified Phase I for the MFRSP is formulated as

min objective (1)

subject to:

constraints (2)-(12), (44), (49)-(51)

$$0 \leq U_i, U_{avg}, B_{ii'} \leq 1 \qquad \forall i, i' \in I \tag{52}$$

Similarly, the Phase III formulation incorporating the Gini-based equity constraint is expressed as:

min objective (30)

subject to:

constraints (31)-(39), (45), (49)-(52)

4.4.2. Equity Consideration with a constraint that ensures a minimum utility threshold

According to Rawls [47], inequality should exist only to the extent that it improves the welfare of the least advantaged. This principle forms the basis of the Rawlsian Maximin Criterion, one of the most prominent fairness measures in operations research. Several studies, including Luss [48], Ogryczak and Śliwiński [49], and Ogryczak et al. [50] have applied the Rawlsian Maximin approach to ensure equitable resource allocation.

The Rawlsian Maximin criterion aims to maximize the minimum utility among all groups within the social welfare function $W(U)$, as expressed in equation (53).

$$W(U) = \min_{i \in I} U_i \tag{53}$$

To apply this principle within our model, we introduce a new parameter u_{min} representing the minimum acceptable utility level across all groups. The following constraint (54) is then added to the model to ensure that the utility of each group meets or exceeds this threshold:

$$U_i \geq u_{min} \qquad \forall i \in I \tag{54}$$

In this approach, the modified Phase I formulation of the MFRSP is expressed as:

min objective (1)

subject to:

constraints (2)-(12), (44), (54)

$$0 \leq U_i \leq 1 \qquad \forall i \in I \tag{55}$$

The modified Phase III formulation incorporating the minimum utility threshold is defined as:

min objective (30)

subject to:

constraints (31)-(39), (45), (54)-(55)

4.4.3. Equity consideration with a constraint that bounds the D-Index

The dissimilarity index (D-Index), proposed by Barros et al. [51], measures the degree to which access rates to a particular service differ among groups distinguished by circumstantial characteristics, such as location, income, or gender, relative to the overall average access rate of the entire population. Both United Nations ESCAP [52] and Abras et al. [53] employ the D-Index to evaluate equity in the distribution of services and opportunities.

Similar to the Gini Index, the D-index ranges between 0 and 1, where 0 indicates perfect equality, and 1 signifies complete inequality, i.e., access to a service is entirely monopolized by a specific group. While the Gini Index measures inequality through pairwise differences in utility values across groups, the D-Index evaluates inequality based on the deviations of each group's utility from the population average. Additionally, the D-Index takes into account the relative size of each group by weighting deviations according to their population proportions.

We introduce a parameter D to represent the upper bound on acceptable inequity among the groups. To ensure that inequity does not exceed this predefined threshold, we incorporate constraint (56) in the model. The constraint enforces that inequity across groups remains within the acceptable limit specified by D.

$$\frac{1}{2U_{avg}} \sum_{i \in I} \epsilon_i |U_{avg} - U_i| \leq D \tag{56}$$

Here, ϵ_i denotes the proportion of group $i \in I$ within the population, that is,

$$\epsilon_i = \frac{\sum_{s \in S} p_s n_{is}}{\sum_{s \in S} \sum_{j \in I} p_s n_{js}}.$$

To linearize constraint (56), we introduce an auxiliary variable F_i, representing the absolute deviation of the utility of group i from the average utility, U_{avg}. Instead of directly using Constraint (56), the following linear constraints (57)-(59) are incorporated to preserve model linearity.

$$\sum_{i \in I} \epsilon_i F_i \leq 2\ D\ U_{avg} \tag{57}$$

$$U_i - U_{avg} \leq F_i \qquad \forall i \in I \tag{58}$$

$$U_{avg} - U_i \leq F_i \qquad \forall i \in I \tag{59}$$

In this approach, the modified Phase I formulation of the MFRSP is expressed as:

$$\begin{aligned}
&\min \quad \text{objective (1)} \\
&\textit{subject to:} \\
&\text{constraints (2)-(12), (44), (57)-(59)} \\
&0 \leq U_i, U_{avg}, F_i \leq 1 \qquad \forall i \in I
\end{aligned} \tag{60}$$

The modified Phase III formulation incorporating the D-Index-based equity constraint is defined as:

$$\begin{aligned}
&\min \quad \text{objective (30)} \\
&\textit{subject to:} \\
&\text{constraints (31)-(39), (45), (57)-(60)}
\end{aligned}$$

4.5. Interrelation of Phases and Overall Solution Framework

This subsection integrates the three components of the proposed methodology into a unified framework, explaining how the three phases, namely, tactical planning, routing, and operational scheduling, interact to form the complete solution process for MFRSP. The interrelation of these phases and the overall solution process is illustrated in Figure 2.

As outlined earlier, **Phase I** performs tactical-level planning by determining how many individuals from each group will receive vaccination services in each VP, without assigning specific days. In the baseline setting (without equity constraints), Phase I always yields a feasible solution. However, when equity considerations are incorporated, feasibility depends on the chosen levels of fairness specified through parameters such as ψ, which sets an upper bound on the Gini Index to limit inequity among groups; $u_{\min}$, which enforces a minimum acceptable utility level for all groups; and D, which bounds the D-Index to control inequity relative to the population-weighted average utility. Depending on these parameter values, Phase I may become infeasible under the existing capacity and/or maximum allowable latency. In such cases, feasibility can be restored by (i) increasing capacity (e.g., by deploying additional MFs) or (ii) relaxing either the equity or lateness constraints.

Once a feasible solution is obtained in Phase I, the solution methodology proceeds to **Phase II**, which focuses on route construction. Using the location-level vaccination quantities determined in Phase I, Phase II designs the daily routes for each MF within each VP, ensuring that all selected locations are visited in accordance with logistical and capacity constraints.

Phase III then refines these tactical and routing-level plans into an operational-level schedule. It generates a detailed daily vaccination plan that specifies, for each MF and each day, how many individuals from each group will receive vaccination service, ensuring consistency with the routes obtained in Phase II. Similar to Phase I, Phase III is always feasible in the absence of equity considerations. However, when equity constraints are imposed, infeasibility may arise due to the simplified travel-time approximation ($\bar{d}$) used in Phase I. Because this approximation may underestimate actual travel times, Phase I may produce a seemingly feasible solution that violates time, capacity, or equity constraints

when exact travel times are enforced in Phase III. To address this, whenever infeasibility occurs in Phase III, Phase I is re-executed using an updated traveling time metric $\bar{d}$, adjusted according to $\bar{d} \leftarrow \bar{d} + \Delta$, where $\Delta = 0.25\sigma$ in our experiments, and σ denotes the standard deviation of d_{jl} for $j, l \in \bigcup_{m \in M} L_m$, with $j \neq l$. Phases II and III are rerun using this revised parameter. This iterative process continues until a feasible solution is achieved in Phase III, at which point the algorithm terminates.

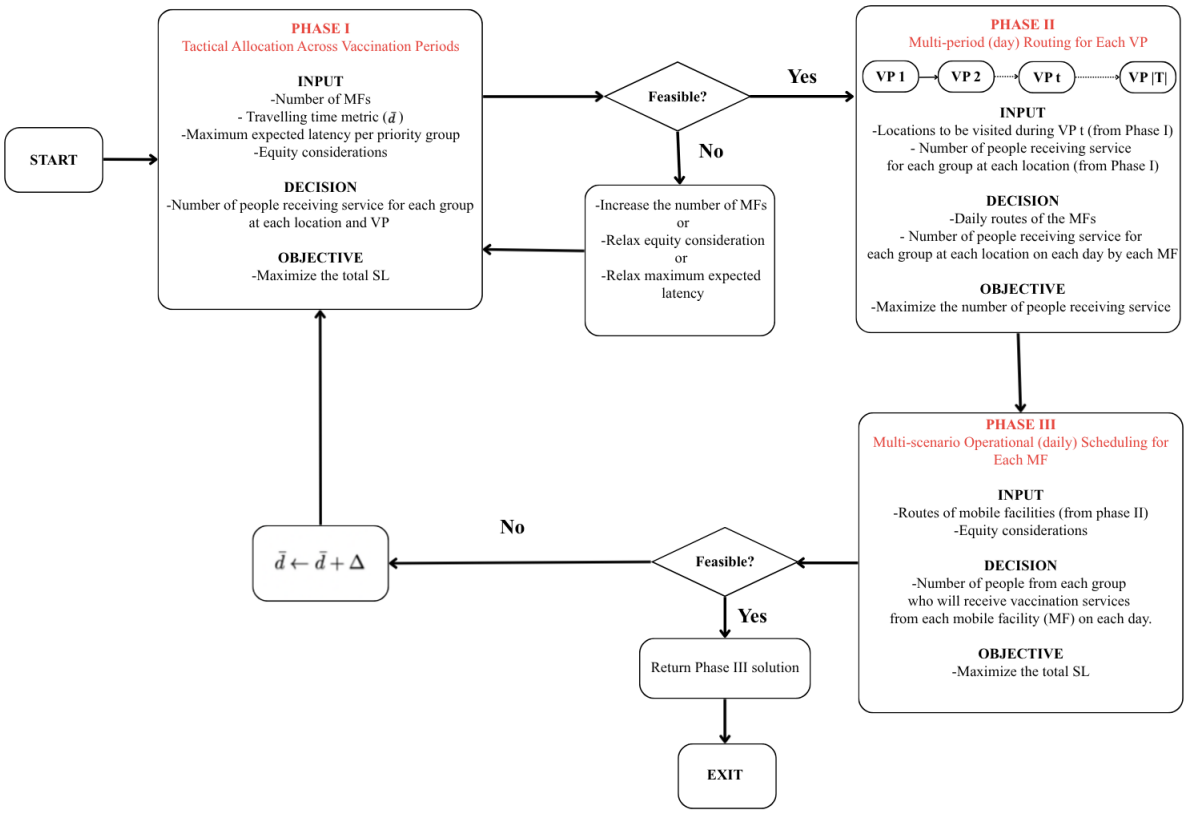


Figure 2: Three-phase solution framework illustrating how tactical allocation (Phase I), routing and scheduling (Phase II), and scenario-based operational planning (Phase III) jointly generate feasible and equity-aware vaccination service plans.

5. Computational Study

This section presents the results of our computational experiments, which were conducted to evaluate the proposed three-phase solution methodology for the MFRSP. The Gaziantep case study represents a large-scale real-world setting, comprising 225 locations, over 3,000 groups, and a 91-day planning horizon. To validate and test the performance of the proposed methodology before applying it to this large-scale problem, we first generate and analyze smaller problem instances. These preliminary experiments serve two purposes: (i) to ensure that the model structure and solution logic function correctly under various configurations, and (ii) to confirm that all constraints and assumptions are consistently satisfied. To assess the impact of equity integration, we evaluate three alternative equity formulations—namely, the *Gini Index*, the *Minimum Utility Threshold*, and the *D-Index*. Their respective performances in terms of service-level outcomes are first compared using the small test instances. The most effective approach is then applied to the full-scale Gaziantep case.

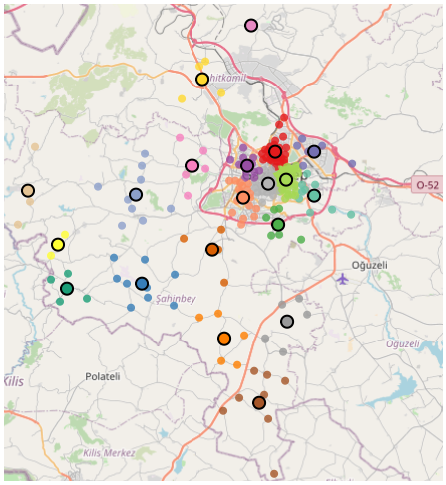


Figure 3: Twenty clusters created using the k-means algorithm

5.1. Computational Experiments with Small Instances

To construct small-sized problem instances, we employ the K-means clustering algorithm (Hartigan and Wong [54]), which partitions the 225 neighborhood locations into clusters by minimizing the within-cluster sum of squared distances between each location and its cluster centroid. A total of 20 clusters are created, with each cluster representing a group of geographically compact regions. Figure 3 shows the resulting 20 clusters, their centroids, and the spatial distribution of locations within each cluster.

For each cluster, we record the centroid coordinates and the assignment of each location to its corresponding cluster. The demand of groups belonging to the same cluster, IVP, and vulnerability category is aggregated to reduce model complexity. To further reduce the problem size, the planning horizon is shortened from 12 to 4 weeks, with MFs assumed to operate three days per week. As a result, the small test instances contain 20 locations, 139 groups, and 12 days within the planning horizon.

To maintain temporal realism, the spillover effect described in Section 4.5 is also incorporated. Specifically, a portion of unserved demand from the previous planning horizon is included in the current one, resulting in an increase in the total number of groups from 139 to 153. In parallel, the planning horizon is extended by one additional VP (resulting in a 5-week scheduling window), which serves as a buffer for deferred demand.

5.1.1. Instance Generation and Computational Complexity

Multiple problem instances are generated by varying the number of MFs ($|M| = 1, 2, 3$) and by applying alternative equity approaches. Table 7 lists these instances: Instances 1–13 correspond to one MF, Instances 14–26 to two MFs, and Instance 27 to three MFs. The second column of Table 7 specifies the type of equity approach employed, and the third column lists the value of the associated equity parameter. When three MFs are available, all demand points can be served even without incorporating equity considerations; therefore, equity-based formulations are only tested for configurations with up to three MFs. For each MF configuration, instances are created both without equity and with one of the three alternative equity formulations: *Gini Index*, the *Minimum Utility Threshold*, and the *D-Index*, each evaluated under multiple equity parameter settings.

As discussed in Section 4.5, Phase I may become infeasible under certain combinations of equity levels, MF capacity limits, or maximum allowable latency. When infeasibility occurs, as illustrated in Figure 2, several recovery strategies can be applied to restore feasibility, such as increasing capacity, relaxing latency constraints, or easing equity requirements. In this study, we adopt the latter approach—relaxing the equity level. As summarized in Table 7, Phase I becomes infeasible for Instances 4, 8, and 13 (one MF) and Instances 18, 22, and 27 (two MFs), all corresponding to stringent equity settings.

All instances are solved using Gurobi 12.0.2. Phase I is solved either to proven optimality or until a time limit of 600 seconds is reached. For each VP $t \in T$, Phase II is solved until the relative optimality gap, defined as $\text{Gap}(\%) = 100(\text{UB} - \text{LB})/\text{LB}$, falls below 1 %, or until a time limit of 600 seconds is reached. Phase III, being a linear program, always reaches to optimality within seconds. However, incorporating equity constraints in Phase I considerably increases model complexity. Similarly, since Phase II represents a vehicle routing model, even moderately sized instances can be computationally demanding. To enhance computational performance, we employ Gurobi's *No-Relaxation (NoRel) Heuristic*, which focuses on finding high-quality feasible solutions prior to solving the LP relaxation [55, 56].

To determine the most effective time allocation for the NoRel heuristic, we used a set of instances, and a total time limit of 600 seconds, while varying the NoRel time across 60, 180, 300, 420, and 540 seconds. In all runs, Phases I and III achieved optimality within seconds, while Phase II exhibited the best performance when the NoRel time was set to 60 seconds. Consequently, all subsequent computational experiments are conducted using a 60-second NoRel time and a total time limit of 600 seconds per phase.

Table 7 reports the relative optimality gaps obtained for Phases I and II. For Phase I, the reported value represents the final gap after the time limit. For Phase II, the table shows whether the relative gap falls below 1%; otherwise, the achieved value is reported for each VP. The results indicate that all instances without equity considerations and those employing the Minimum Utility Threshold achieve optimality in Phase I. However, some instances incorporating the Gini Index or D-Index remain unsolved within the time limit due to the additional nonlinear terms and inter-group utility comparisons introduced by these formulations.

The computational complexity introduced by each equity mechanism can be quantified by the number of new variables and constraints added in Phase I. When equity is enforced via the *Minimum Utility Threshold*, one decision variable U_i and two additional constraints [(44), (54)] are introduced per group $i \in I$, adding a total of $|I|$ variables and $2|I|$ constraints. When the *Gini Index* is used, variables U_i and $B_{ii'}$ are introduced in the order of $|I|^2$, along with constraints [(44), (49)–(51)], adding approximately $2|I|^2 + |I| + 1$ new constraints. For the *D-Index*, variables U_i, U_{avg}, and F_i are introduced in the order of $|I| + 1$, together with constraints [(44), (57)–(59)], contributing about $3|I| + 1$ new constraints. These additional components significantly increase the model's size and solution time, particularly under higher equity levels.

This growing complexity is reflected in the observed infeasibilities and gaps. For instance, Phase III

Table 7: Instances and their relative optimality gaps, Gap(%) = 100(UB − LB)/LB, for Phases I and II after the time limit. A "✓" indicates that a relative gap below 1% was reached before the time limit, while numeric values denote the final gap (%). Instance 21†: Phase III infeasible, and Phase I becomes infeasible when $\bar{d}$ is increased by Δ, where $\Delta = 0.25\sigma$.

Instance	**No. of MFs**	**Equity Approach**	**Equity Parameter**	**Phase I Gap(%)**	**Phase II Gap(%) per VP**				
					VP1	VP2	VP3	VP4	VP5
1	1	None	–	✓	✓	✓	✓	✓	✓
2	1	Gini Index	$\psi = 0.4$	✓	✓	✓	✓	1.85	✓
3	1	Gini Index	$\psi = 0.3$	2.17	✓	4.53	✓	✓	✓
4	1	Gini Index	$\psi = 0.2$	Inf.	–	–	–	–	–
5	1	Min. Utility	$u_{\min} = 0.1$	✓	✓	✓	✓	✓	✓
6	1	Min. Utility	$u_{\min} = 0.2$	✓	✓	✓	✓	✓	✓
7	1	Min. Utility	$u_{\min} = 0.3$	✓	✓	✓	✓	✓	✓
8	1	Min. Utility	$u_{\min} = 0.4$	Inf.	–	–	–	–	–
9	1	D-Index	$D = 0.1$	✓	✓	✓	✓	✓	2.78
10	1	D-Index	$D = 0.05$	3.56	✓	✓	✓	✓	✓
11	1	D-Index	$D = 0.025$	8.23	✓	1.4	✓	✓	✓
12	1	D-Index	$D = 0.01$	27.32	3.85	5.26	11.72	✓	✓
13	1	D-Index	$D = 0.005$	Inf.	–	–	–	–	–
14	2	None	–	✓	1.78	1.24	1.01	1.87	✓
15	2	Gini Index	$\psi = 0.2$	✓	2.14	1.52	2.27	✓	✓
16	2	Gini Index	$\psi = 0.1$	✓	2.66	1.03	1.47	1.17	✓
17	2	Gini Index	$\psi = 0.05$	1.59	3.16	✓	1.39	1.99	✓
18	2	Gini Index	$\psi = 0.025$	Inf.	–	–	–	–	–
19	2	Min. Utility	$u_{\min} = 0.5$	✓	3.25	✓	✓	✓	✓
20	2	Min. Utility	$u_{\min} = 0.6$	✓	2.82	✓	✓	✓	✓
21	2	Min. Utility	$u_{\min} = 0.7$	†	–	–	–	–	–
22	2	D-Index	$D = 0.01$	✓	2.67	✓	✓	✓	✓
23	2	D-Index	$D = 0.005$	✓	2.83	✓	✓	✓	✓
24	2	D-Index	$D = 0.0025$	0.73	2.33	✓	✓	✓	✓
25	2	D-Index	$D = 0.001$	1.28	3.22	✓	1.18	✓	✓
26	2	D-Index	$D = 0.0005$	Inf.	–	–	–	–	–
27	3	None	–	✓	✓	✓	✓	✓	✓

infeasibility in Instance 21 indicates that the travel-time approximation $\bar{d}$ used in Phase I is underestimated. When the travel-time metric is updated as $\bar{d} \leftarrow \bar{d} + \Delta$ (with $\Delta = 0.25\sigma$, where σ denotes the standard deviation of d_{jl} for $j, l \in \bigcup_{m \in M} L_m$, $j \neq l$), Phase I becomes infeasible. This outcome exemplifies the increased computational complexity and tighter feasibility margins introduced by the *Minimum Utility Threshold* formulation at higher parameter levels. The same trend is evident in Instance 12 (D-Index, $D = 0.01$), which exhibits a 27.32% optimality gap in Phase I, further highlighting the trade-off between equity enforcement and computational tractability.

5.1.2. Analysis of Service Levels and Utility of Groups

The Phase III results are presented in Table 8. Instances 4, 8, 13, 18, 21, and 26—where Phase I is infeasible—are omitted for brevity. The remaining results demonstrate how varying the number of MFs and the degree of equity enforcement influences the overall objective values and service-level distributions across vulnerable and non-vulnerable groups.

For the scenarios with a single MF, the objective function value (i.e., the weighted total service level) is 1.85 when no equity is enforced, with 106 out of 153 groups served either fully or partially.

Table 8: Objective function values and service levels achieved in Phase III with small instances. Instances with Phase I infeasibility (#4, #8, #13, #18, #21, #26) are omitted from the table for brevity.

Instance	**MFs**	**Equity Approach**	**Parameter**	**Objective Value** ($\eta^V Q^V + \eta^{NV} Q^{NV}$)	Q^V	Q^{NV}	**No. of Groups Served** (Full/Partial)	**Utility**		
								Min	**Avg.**	**Max**
1	1	None	–	1.85	0.49	0.87	106	0.00	0.57	1.00
2	1	Gini Index	$\psi = 0.4$	1.85	0.49	0.87	106	0.00	0.57	1.00
3	1	Gini Index	$\psi = 0.3$	1.82	0.46	0.90	122	0.00	0.66	1.00
5	1	Min. Utility	$u_{\min} = 0.1$	1.71	0.44	0.83	153	0.10	0.48	1.00
6	1	Min. Utility	$u_{\min} = 0.2$	1.67	0.42	0.83	153	0.20	0.51	1.00
7	1	Min. Utility	$u_{\min} = 0.3$	1.63	0.39	0.85	153	0.30	0.57	1.00
9	1	D-Index	$D = 0.10$	1.68	0.47	0.74	117	0.00	0.50	1.00
10	1	D-Index	$D = 0.05$	1.57	0.47	0.62	128	0.00	0.45	1.00
11	1	D-Index	$D = 0.025$	1.42	0.45	0.51	133	0.00	0.42	1.00
12	1	D-Index	$D = 0.01$	1.23	0.39	0.45	151	0.00	0.36	0.97
14	2	None	–	2.84	0.93	0.99	146	0.00	0.82	1.00
15	2	Gini Index	$\psi = 0.2$	2.84	0.93	0.99	150	0.00	0.82	1.00
16	2	Gini Index	$\psi = 0.1$	2.84	0.93	0.99	151	0.00	0.87	1.00
17	2	Gini Index	$\psi = 0.05$	2.79	0.90	0.98	153	0.22	0.88	1.00
19	2	Min. Utility	$u_{\min} = 0.5$	2.83	0.91	1.00	153	0.50	0.87	1.00
20	2	Min. Utility	$u_{\min} = 0.6$	2.82	0.91	1.00	153	0.60	0.89	1.00
22	2	D-Index	$D = 0.01$	2.74	0.88	0.97	152	0.00	0.86	1.00
23	2	D-Index	$D = 0.005$	2.68	0.88	0.93	153	0.43	0.85	1.00
24	2	D-Index	$D = 0.0025$	2.64	0.87	0.90	153	0.42	0.84	1.00
25	2	D-Index	$D = 0.001$	2.11	0.70	0.72	153	0.55	0.66	1.00
27	3	None	–	3.00	1.00	1.00	153	1.00	1.00	1.00

Incorporating equity through the *Gini Index* with $\psi = 0.4$ yields the same result, indicating that this equity level is insufficient to influence the allocation. Reducing ψ to 0.3 results in a minor decrease in the objective value (to 1.82) but expands the number of groups served to 122, indicating a modest improvement in equity.

Under the *Minimum Utility Threshold* approach, increasing $u_{\min}$ from 0.1 to 0.3 slightly reduces the overall objective value but ensures that all 153 groups receive at least partial service. This method directly enforces a baseline service level, thereby enhancing fairness without compromising efficiency excessively. In contrast, the *D-Index* approach produces lower objective values and incomplete coverage, as its fairness adjustments shift resources toward groups with lower initial access at the expense of overall efficiency.

Overall, the highest total service level is achieved without considering equity, as MFs prioritize geographically proximate locations and maximize service within shorter travel times. Once equity is imposed, MFs must travel to more dispersed locations, which reduces the achievable service volume but improves fairness. Figure 4 illustrates the percentage of groups served (fully or partially) for each equity approach when one MF is deployed, highlighting that the *Minimum Utility Threshold* approach ensures full or partial coverage across all groups.

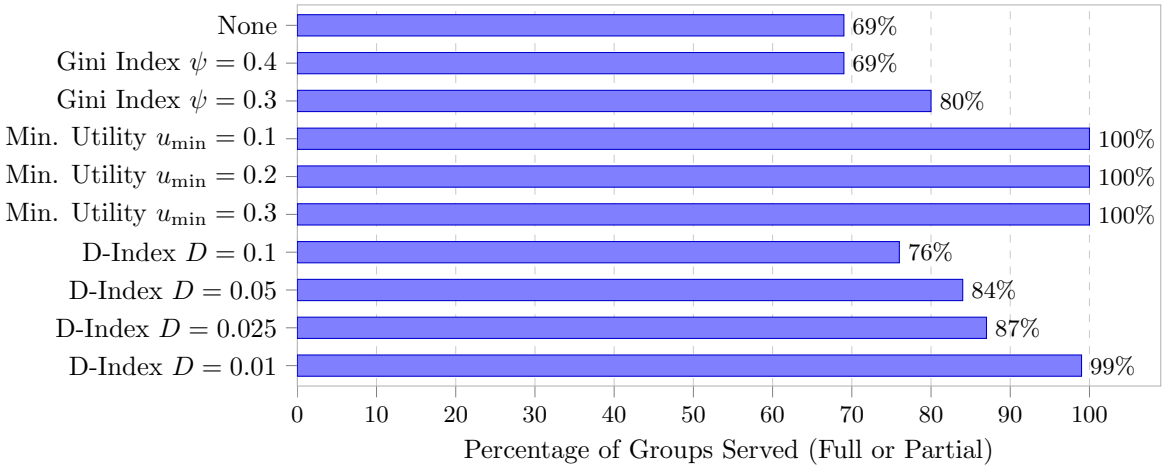


Figure 4: Comparison of percentage of groups receiving full or partial service under different equity approaches (1 MF case).

For the two-MF scenarios, similar patterns are observed. With higher overall capacity, full coverage can be achieved under both the *Gini Index* (Instance 17) and the *D-Index* (Instances 23–24), as well as under the *Minimum Utility Threshold*. Nevertheless, the corresponding objective function values for the *Gini Index* and *D-Index* remain lower than those for the *Minimum Utility Threshold* (Instances 19–20), which balances fairness and efficiency more effectively. Finally, when three MFs are available, as noted before, full coverage is achieved even without equity enforcement. These findings indicate that the *Minimum Utility Threshold* approach offers the most balanced trade-off between fairness, service coverage, and computational efficiency. Consequently, this formulation is selected for the real-world Gaziantep case study as the most appropriate method for ensuring equitable and effective vaccination service delivery.

5.1.3. Analysis of Priority Levels Assigned to Vulnerable and Non-Vulnerable Groups

In the computational experiments, priority is assigned to vulnerable groups by setting $\eta^V = 2$ and $\eta^{NV} = 1$, giving vulnerable groups twice the weight of non-vulnerable ones. However, as shown in Table 8, the percentage of non-vulnerable groups served remains higher across all instances, despite this prioritization. This result primarily stems from the structural characteristics of the dataset: vulnerable groups are roughly four times that of non-vulnerable groups and are also tend to have larger population sizes. Thus, when a MF visits a location, it typically serves both group types; but, the smaller non-vulnerable groups are easier to cover fully, yielding higher aggregate service levels for them.

To further examine the effect of priority weighting, we analyze a representative instance—one MF with $u_{\min} = 0.3$, Instance 7—under several η^V and η^{NV} configurations. The resulting efficient frontier is shown in Figure 5. With equal weights ($\eta^V = \eta^{NV} = 1$), the service levels are 0.34 for vulnerable groups and 0.90 for non-vulnerable groups. Increasing the priority to $\eta^V = 2$ (keeping $\eta^{NV} = 1$) improves the service level of vulnerable groups by 14% while reducing that of non-vulnerable groups by 6%, resulting in values of 0.39 and 0.85, respectively. Further increasing η^V to 3 yields an 8% gain for vulnerable groups but a 15% reduction for non-vulnerable ones. With $\eta^V = 4$, the marginal gain

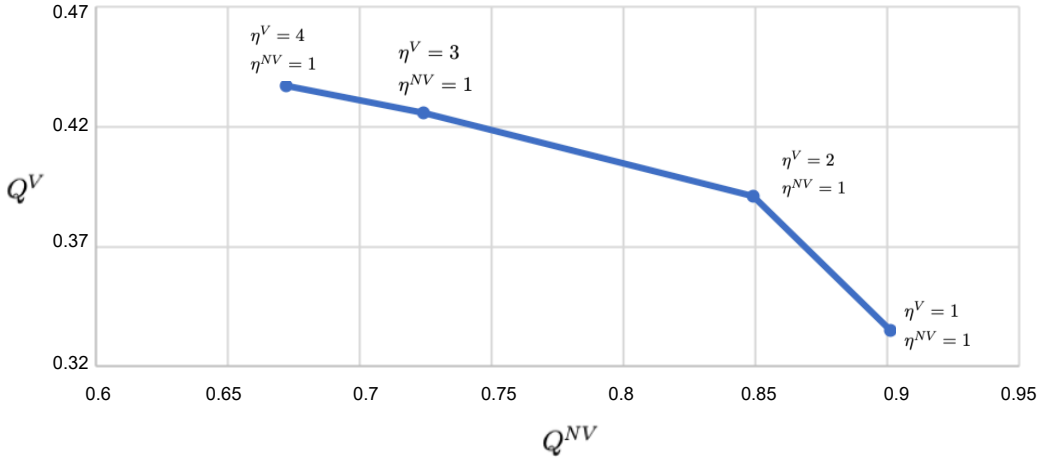


Figure 5: Trade-off between service levels of vulnerable (Q^V) and non-vulnerable (Q^{NV}) groups for different values of the priority parameters η^V and η^{NV}.

for vulnerable groups drops to only a 2.5%, while the service level for non-vulnerable groups declines by 7%.

As is typical in multi-objective settings, improvements for one group come at the cost of the other. For $\eta^V = 3$ and $\eta^V = 4$, the reduction in service level for non-vulnerable groups outweighs the incremental gains for vulnerable ones. Therefore, the configuration $\eta^V = 2$ and $\eta^{NV} = 1$ offers the most balanced trade-off between equity and efficiency and is adopted for all subsequent analyses.

6. Case Study: Vaccination of People in the Gaziantep Province of Türkiye

Syrian refugees have been residing in Türkiye since 2011, when the Syrian civil war began. According to the data of the Presidency of Migration Management (as of May 1, 2025), Istanbul hosts the largest Syrian population in Türkiye, with approximately 481,000 people, followed by Gaziantep, located near the Syrian border, with around 385,000 people [6]. In Gaziantep, the districts of Şahinbey and Şehitkamil accommodate the majority of this population [57]. The health directorate of the province deployed MFs during the vaccination campaign. Here, we focus on designing optimized mobile COVID-19 vaccination services for both Syrian refugees and Turkish citizens residing in these two districts. Data obtained from the Gaziantep Municipality indicate that the vaccination rate among refugees is significantly lower than that of Turkish citizens. Although multiple factors may contribute to this disparity, limited accessibility to vaccination centers is considered one of the main challenges. To address this issue, we assume that deploying mobile vaccination facilities closer to refugee-populated neighborhoods could improve vaccine accessibility and overall coverage. Accordingly, we consider providing COVID-19 vaccination services to all refugees and to selected Turkish citizens who have difficulty accessing fixed health centers over a 12-week planning horizon.

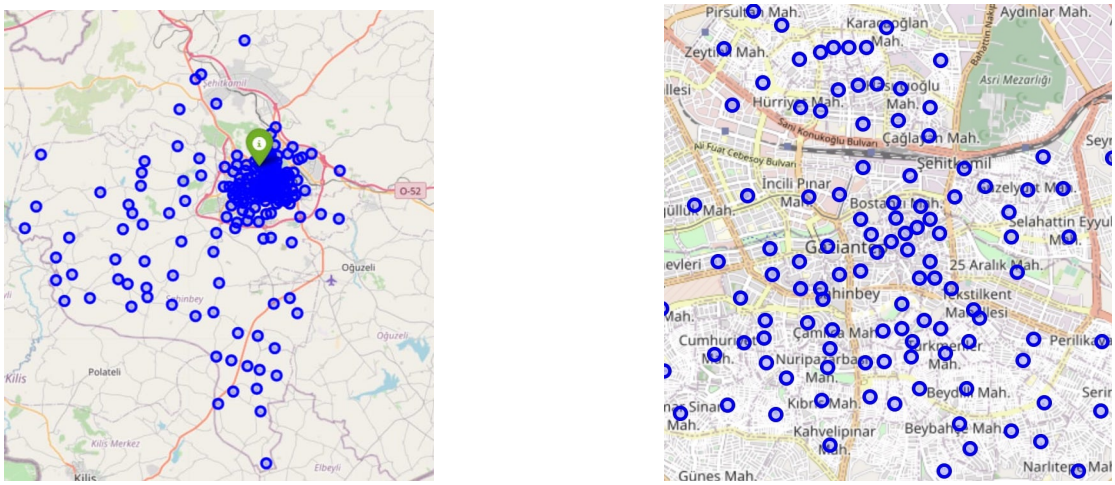
Turkish citizens typically receive healthcare through family health centers, which are located in almost every neighborhood. Hence, vaccination services via MFs are targeted only toward Turkish citizens who live more than 1 km away from the nearest health facility. MFs are assumed to operate seven

days a week and provide daily vaccination services according to the optimized schedules obtained from our model.

6.1. Data Preparation

The Gaziantep Municipality supplied demographic data on the number of Turkish citizens and Syrian refugees in each of the 225 neighborhoods across Şahinbey and Şehitkamil districts. The combined population of these neighborhoods is $1,626,328$ Turkish citizens and $329,688$ refugees. Vaccination services are planned for all refugees in these 225 neighborhoods and for Turkish citizens in the 55 neighborhoods that lack a health institution within a 1 km radius.

At the beginning of each day, all MFs collect the required vaccine doses from the Gaziantep Provincial Health Directorate, which serves as the depot and point of departure. Figure 6a shows all 225 neighborhoods along with the depot location, i.e., the Gaziantep provincial health department (marked in green), while Figure 6b presents a zoomed-in view of the central area. The dataset used in this study is publicly available at [58], enabling further use by researchers and policymakers.



(a) Neighborhoods in which vaccination service will be provided

(b) Zoomed-in view of central neighborhoods

Figure 6: Spatial distribution of the 225 neighborhoods included in the Gaziantep vaccination case study (left) and a zoomed-in view of the central high-density region (right)

Vaccination is planned for individuals aged 12 years and older. According to Refugees Association [6], 37% of Syrian refugees, and according to data from Turkish Statistical Institute [59], 19% of Turkish citizens, fall within the 0–12 age group. Based on these figures, 207,752 refugees and 24,364 Turkish citizens are eligible for vaccination in our study.

To reflect vaccine hesitancy due to factors such as cultural considerations, individual preferences, and misinformation, three scenarios are generated to represent high, medium, and low hesitancy levels, where the medium hesitancy level (baseline) corresponds to having 50% of the refugees and 60% of the Turkish citizens willing to receive the vaccine. In the high hesitancy level case, 40% of the refugees and 50% of the Turkish citizens are willing, and in the low hesitancy level case, 60% of the refugees and 70% of the Turkish citizens are willing. The values for medium hesitance are consistent with vaccination completion rates reported by the Gaziantep Municipality (50% among refugees) and by

Mathieu et al. [60] (61.08% among Turkish citizens). Each scenario is assumed to have an equal probability of occurrence, i.e., $p_s = 1/3$ for any scenario s.

The vaccination schedule is planned over a 12-week horizon, consistent with the fact that booster doses for COVID-19 vaccines are not administered until at least 3 months after the initial vaccination. Each individual is randomly assigned an IVP between weeks 1 and 12, with equal probability. Groups are formed based on individuals with the same location, IVP, and vulnerability category. Two vulnerability categories are defined: (i) Vulnerable groups (I^V): Syrian refugees, due to language barriers and limited access to healthcare services, and (ii) Non-vulnerable groups (I^{NV}): Turkish citizens, who generally face fewer accessibility constraints.

The resulting dataset comprises 2,379 vulnerable (refugee) and 660 non-vulnerable (Turkish citizen) groups, totaling 3,039 groups. To prioritize vulnerable groups, the model uses η^V=2 and η^{NV}=1. Similarly, the maximum lateness thresholds are set to γ^V=0.2 and γ^{NV}=0.3, thereby imposing a lower expected total latency on vulnerable groups.

To model spillover effects, approximately one-twelfth of the total population from the previous planning horizon (equivalent to a single IVP) is carried into the current horizon, increasing the total number of groups to 3,292. The planning horizon is extended to 13 weeks, where the 13th week serves as a buffer period rather than an assigned IVP. This buffer allows limited postponement of vaccinations beyond the 12th week, preventing artificial congestion of demand near the end of the planning horizon and ensuring smoother continuity between consecutive horizons.

Facility capacities are considered in two dimensions: the number of vaccine doses that an MF can administer daily and the total operational time available per day. In the computational experiments, each mobile facility (MF) is assumed to carry up to 500 vaccine doses and to operate for 8 hours (including travel and service time) per day. We assume an average travel speed of 1 km per minute, a 30-minute setup time required per location before vaccination begins, and a three-minute vaccination time. Every MF is staffed by three healthcare personnel, enabling the team to vaccinate, on average, one person per minute over the course of the day. These assumptions together provide a realistic representation of the operational capabilities and resource limitations of mobile vaccination units in the field.

6.2. Computational Experiments with the Gaziantep Case

In the Gaziantep case, we consider a large-scale setting with 225 locations and more than 3,000 groups. We generate a total of 17 test instances, labeled G1–G17, where each instance corresponds to a specific combination of fleet size and equity requirement. The fleet size takes values $|M| \in \{3, 4, 5, 6\}$, and equity is imposed via the *Minimum Utility Threshold* at different equity levels $u_{\min}$. We begin the experiments with three MFs and gradually increase the fleet size to six MFs, which is sufficient to serve all groups, even without considering equity.

All instances are solved using Gurobi 12.0.2. Phase I is solved either to proven optimality or until a time limit of 1,800 seconds is reached. For each VP $t \in T$, Phase II is solved until the relative optimality gap, computed as $\text{Gap}(\%) = 100(\text{UB} - \text{LB})/\text{LB}$, falls below 1%, or until the same time limit of 1800

seconds is reached. Phase III is a linear program and always attains optimality; however, the high dimensionality of the Gaziantep case leads to non-negligible initialization and presolve times for Gurobi before the simplex or barrier steps begin. As in the small-instance experiments, we employ Gurobi's *No-Relaxation (NoRel) heuristic* to enhance computational performance. To calibrate the NoRel time, we run a set of representative Gaziantep instances with a total time limit of 1,800 seconds and vary the NoRel allocation across 300, 600, 900, 1,200, and 1,500 seconds. In all tests, Phases I and III reach optimality, whereas Phase II yields the tightest optimality gaps when NoRel is allocated 1,500 seconds, consistently producing better bounds for all VPs. Therefore, in all subsequent experiments, we adopt a NoRel time of 1,500 seconds within a total time limit of 1,800 seconds per phase.

Table 9: Objective function values and service levels achieved in Phase III for the Gaziantep case. Instances marked with † indicate Phase I infeasibility.

Instance	**MFs**	**Equity Approach**	**Parameter**	**No. of $\bar{d}$ Increments**	**Objective Value** ($\eta^V Q^V + \eta^{NV} Q^{NV}$)	Q^V	Q^{NV}	**Groups Served** (Full/Partial)	**Utility**		
									Min	**Avg.**	**Max**
G1	3	None	–	–	2.28	0.65	0.98	2283	0.00	0.51	1.00
G2	3	Min. Utility	$u_{\min} = 0.1$	–	2.20	0.61	0.99	3292	0.10	0.54	1.00
G3	3	Min. Utility	$u_{\min} = 0.2$	1	2.16	0.58	1.00	3292	0.20	0.58	1.00
G4	3	Min. Utility†	$u_{\min} = 0.3$	1	–	–	–	–	–	–	–
G5	3	Min. Utility†	$u_{\min} = 0.4$	–	–	–	–	–	–	–	–
G6	4	None	–	–	2.69	0.85	0.99	2792	0.00	0.70	1.00
G7	4	Min. Utility	$u_{\min} = 0.3$	–	2.65	0.83	0.99	3292	0.30	0.75	1.00
G8	4	Min. Utility	$u_{\min} = 0.4$	1	2.61	0.81	0.99	3292	0.40	0.74	1.00
G9	4	Min. Utility	$u_{\min} = 0.5$	2	2.56	0.78	1.00	3292	0.50	0.75	1.00
G10	4	Min. Utility†	$u_{\min} = 0.6$	–	–	–	–	–	–	–	–
G11	5	None	–	–	2.93	0.97	1.00	3166	0.00	0.87	1.00
G12	5	Min. Utility	$u_{\min} = 0.5$	–	2.93	0.96	1.00	3292	0.50	0.90	1.00
G13	5	Min. Utility	$u_{\min} = 0.6$	–	2.92	0.96	1.00	3292	0.60	0.91	1.00
G14	5	Min. Utility	$u_{\min} = 0.7$	1	2.90	0.95	1.00	3292	0.70	0.90	1.00
G15	5	Min. Utility†	$u_{\min} = 0.8$	1	–	–	–	–	–	–	–
G16	6	None	–	–	2.99	0.99	1.00	3292	0.22	0.93	1.00
G17	6	Min. Utility	$u_{\min} = 0.9$	2	2.98	0.99	1.00	3292	0.90	0.97	1.00

Table 9 summarizes the Gaziantep instances and their Phase III outcomes. In several cases, the travel-time approximation $\bar{d}$ is updated more than once. This occurs when Phase I is feasible, but Phase III turns out to be infeasible. In such cases, the algorithm increases $\bar{d}$ and restarts the procedure. For instances G3, G4, G8, G9, G14, G15, and G17, feasibility in Phase III is achieved only after one or more $\bar{d}$ updates. In contrast, Phase I is infeasible for G4, G5, G10, and G15, requiring a relaxation of the equity level. In total, we obtain feasible solutions for 13 instances (G1–G3, G6–G9, G11–G14, and G16-G17). For all of these, Phases I and III are solved to optimality. Phase II involves solving a routing model for each of the 13 weeks; across the 13 instances, this yields 169 routing problems. Of these, 109 reach a relative optimality gap below 1% within the time limit, while the remaining 60 have a 2.6% average optimality gap.

The service levels and utilities of the groups are then analyzed for varying mobile fleet sizes. Without equity considerations, 69%, 85%, and 96% of the groups receive full or partial service with three, four, and five MFS, respectively, and all groups were served when six MFs were available. Incorporating equity through the *Minimum Utility Threshold* approach raises the minimum guaranteed service level across all groups, with only modest reductions in the weighted objective value. With three MFs, equity

levels $u_{\min} = 0.1$ and $u_{\min} = 0.2$ ensure full/partial service for all groups with a minor reduction in the weighted service level. With four MFs, equity levels $u_{\min} = 0.3$, 0.4, and 0.5 guarantee full/partial service to all groups. With five MFS, the same holds for $u_{\min} = 0.5$, 0.6, and 0.7. For six MFs, although all groups are already achieve at least 0.22 utility without equity considerations, the *Minimum Utility Threshold* approach with $u_{\min} = 0.9$ raises the minimum service level to 0.9 across all groups. These results imply that, in this setting, the minimum guaranteed utility is approximately 0.2 with three MFs, 0.5 with four MFs, 0.7 with five MFs, and 0.9 with six MFs.

Figure 7 displays the weighted service levels as well as the outcomes for vulnerable and non-vulnerable groups. Instances G1–G3 correspond to 3-MF cases; G6–G9 to 4-MF cases; G11–G14 to 5-MF cases; and G16-G17 to the 6-MF cases. Across all instances, even without equity constraints, at least 98% of non-vulnerable groups receive service, mainly due to their relatively small population sizes, as further discussed in detail in Section 5.1.3. Service levels for vulnerable groups (Q^V) vary depending on both the number of MFs and the imposed minimum utility threshold.

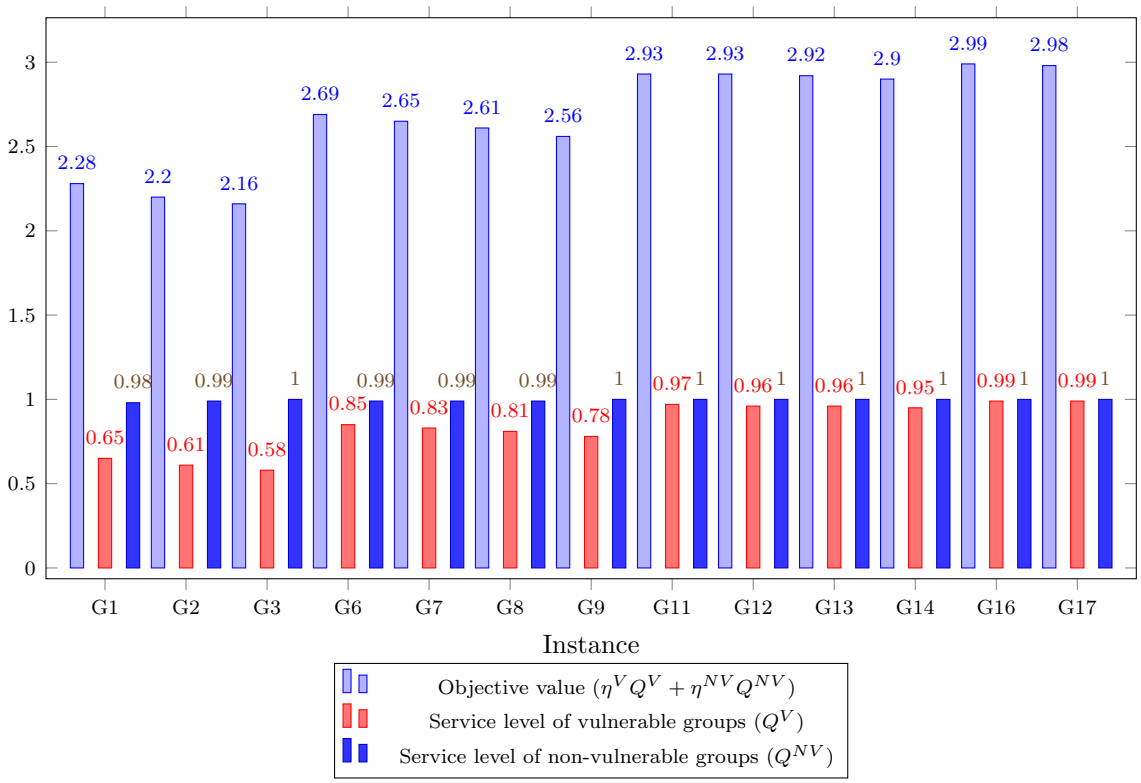


Figure 7: Comparison of total weighted service levels and service levels of vulnerable and non-vulnerable groups for Gaziantep instances. G1–G3: 3 MFs; G6–G9: 4 MFs; G11–G14: 5 MFs; G16-G17: 6 MFs.

With three MFs and without equity consideration, $Q^V = 0.65$. Imposing minimum utility levels of 10% and 20% for all groups reduces Q^V to 0.61 (-6.2%) and 0.58 (-10.8%), respectively. When four MFs exist, without equity consideration, Q^V=0.85. Minimum utility levels of 30%, 40%, and 50% lower Q^V to 0.83 (-2.4%), 0.81 (-4.7%), and 0.78 (-8.2%), respectively. With five MFs and without equity, Q^V=0.97. Imposing a 60% minimum utility level yields Q^V=0.96 (-1%), the same value obtained with the 50% utility threshold, while in both cases $Q^{NV} = 1$. Thus, the 60% equity level dominates the 50% level, providing a stricter equity guarantee with no loss in aggregate performance. When the minimum requirement rises to 70%, Q^V decreases slightly to 0.95 (-2%). With six MFs

and without equity consideration, $Q^V = 0.99$. Imposing a minimum utility level of 90% for all groups leaves Q^V unchanged. These findings highlight a clear trade-off. Enforcing minimum service levels for all groups slightly reduces the aggregate service level for vulnerable groups, but in return guarantees a more balanced and predictable level of protection across the entire population. This provides policymakers with practical guidance on selecting fleet sizes and equity targets that strike a balance between efficiency and fairness.

6.2.1. Managerial Insights from the Gaziantep Case

The Gaziantep case provides several managerial insights for planners responsible for organizing mobile vaccination services, as stated below.

Accurate travel-time approximations are critical for feasibility: A number of instances (G3, G4, G8, G9, G14, G15, G17) required iterative increases in the travel-time approximation $\bar{d}$, showing that rough estimates can cause infeasibility in the routing/scheduling phase even when Phase I allocations are feasible. The simple average pairwise distance used in this study may not fully capture spatial heterogeneity, such as population density patterns or varying travel distances. Developing refined approximation schemes or data-driven travel-time predictors could improve accuracy and reduce late-stage infeasibility.

Small fleet expansions generate disproportionately large coverage gains: Without equity constraints, the proportion of groups receiving full or partial service rises steeply with fleet size: 69% with 3 MFs; 85% with 4; 96% with 5; 100% with 6. This highlights strong scale effects-modest increases in fleet size can substantially expand service coverage, helping planners justify incremental investments in additional mobile units.

Minimum-utility constraints provide strong equity protection with limited efficiency loss: The Minimum Utility Threshold approach enables decision-makers to ensure fairer service distribution that considers disadvantaged areas without sacrificing overall performance. Minimum guaranteed utilities of 0.2 (with 3 MFs), 0.5 (with 4 MFs), 0.7 (with 5 MFs), and 0.9 (with 6 MFs) remain feasible, demonstrating that meaningful equity guarantees can be imposed while preserving high levels of service. This provides decision makers with a predictable and transparent policy tool for public health agencies to balance service fairness and service efficiency.

Equity requirements slightly reduce service for vulnerable groups but improve balance across all groups: Under tighter equity constraints, the service level for vulnerable groups (Q^V) decreases by approximately 6–11% (3 MFs), 2–8% (4 MFs), 1–2% (5 MFs), and 0% (6 MFs). This reflects the expected redistribution effect of equity measures: when resources are scarce, services are reallocated more evenly across population groups. As fleet size grows, this trade-off becomes much smaller, showing that equity becomes cheaper in resource-rich settings.

6.2.2. Interpreting Equity-Aware Mobile Vaccination Planning through the Levesque Access-to-Care Framework

The computational results of the Gaziantep case study can be interpreted through the Levesque Framework of Access to Healthcare [13], providing a structured lens to translate optimization outcomes

into public health–relevant insights. While our model is not explicitly built on the framework ex ante, its outputs align naturally with the framework's multidimensional conception of access, linking operational decisions to utilization-related barriers faced by vulnerable populations.

Approachability: The proposed planning framework enhances Approachability by explicitly targeting 55 neighborhoods without local vaccination facilities, covering 3,292 population clusters. Prior studies emphasize that for immigrant and marginalized populations, services must be proactively brought to communities to become visible and reachable [1]. By systematically identifying underserved locations and deploying mobile facilities, our model addresses the "invisible" groups that may otherwise remain excluded due to geographic isolation or information gaps. The Gaziantep case study results further demonstrate that increasing the fleet size substantially enhances the system's ability to reach underserved populations, thereby reinforcing the role of supply-side outreach in improving access.

Acceptability: Acceptability barriers are reflected through the scenario-based modeling of vaccine uptake. Even when logistical availability is ensured, utilization may remain limited due to trust issues, misinformation, or sociocultural concerns. Consistent with findings by Şimşek et al. [10], lower-than-expected turnout can persist despite high vaccination intent. While the scenario-based structure captures this fragility, the results indicate that equity-oriented prioritization can partially mitigate the adverse effects of low acceptance by safeguarding service levels for vulnerable groups. Nevertheless, acceptability emerges as a critical issue that cannot be resolved solely through logistical optimization, underscoring the need for complementary community engagement strategies.

Availability and Accommodation: The strongest effects of the proposed approach manifest in the availability and accommodation dimensions. Fleet size analyses demonstrate clear improvements in service availability as mobile capacity increases. Accommodation is further strengthened through the integration of IVPs and explicit lateness thresholds, aligning service timing with eligibility and urgency. Prior access studies highlight that limited temporal flexibility is a major barrier for vulnerable populations, whose daily schedules often conflict with fixed service hours [39, 61]. By jointly optimizing spatial deployment and temporal scheduling, our model demonstrates that time is as critical as distance in achieving equitable access.

Affordability: The efficiency–equity trade-off observed in the computational experiments reflects affordability considerations at both system and individual levels. Stronger equity constraints may increase travel distances or render some instances infeasible under tight capacity, indicating higher operational burden. However, the model maintains economic viability by minimizing total travel distance and reducing indirect costs borne by individuals, such as travel time and missed work. The results support existing economic arguments that equitable healthcare delivery can be achieved with controlled efficiency losses when resources are allocated toward populations with the highest unmet need [11, 12].

Appropriateness: Appropriateness is primarily addressed through the Rawlsian *Minimum Utility Threshold* mechanism. By guaranteeing a baseline service level across all neighborhoods—regardless of population size or social status—the model promotes both technical quality (timely service and adequate visit frequency) and interpersonal equity (prioritization of vulnerable groups). This focus

on minimum guaranteed outcomes shifts attention from aggregate volume targets to the fit between service delivery and community needs. As emphasized by Cu et al. [39], such alignment between services and local conditions is central to achieving appropriate access. Our study addresses such a fit through the mathematical guarantee in the form of a minimum service floor for the disadvantaged populations.

Overall, interpreting the computational results through the Levesque Framework highlights how equity-aware routing and scheduling decisions translate into multidimensional access improvements. This perspective strengthens the policy relevance of the proposed model by linking optimization outputs to well-established concepts in health access research.

7. Conclusion

In this study, we examine the Mobile Facility Routing and Scheduling Problem (MFRSP), which concerns planning vaccination services delivered by mobile facilities across multiple periods. The goal is to determine when and where mobile teams should operate to provide adequate capacity, maintain high service quality, and use the resources efficiently. Motivated by disparities in access among vulnerable populations, the MFRSP incorporates prioritization rules, fairness requirements, vaccine hesitancy, and resource limitations within a unified framework to enhance timely and equitable vaccination.

We develop a three-phase matheuristic that links tactical allocation, daily routing, and scenario-based operational scheduling. Phase I assigns vaccination quantities across periods; Phase II converts these allocations into feasible daily routes; and Phase III determines realized vaccination levels under uncertainty. The model maximizes service levels for both vulnerable and non-vulnerable groups, uses group-specific lateness limits to protect high-priority groups, and embeds three equity mechanisms: the Gini Index, a Minimum Utility Threshold, and the D-Index.

To explore model behavior, we first generate small clustered instances to study the effects of equity mechanisms and fleet size. Results show that equity considerations are crucial for balanced service delivery, and the Minimum Utility Threshold consistently provides the most equitable outcomes with minimal loss in efficiency.

The Gaziantep case study demonstrates the framework's scalability, coordinating routing, scheduling, and equity across 225 locations, more than 3,000 population groups, and over 230,000 individuals. Varying fleet size from three to six mobile units reveals clear efficiency–equity trade-offs: coverage improves steadily with additional vehicles, and imposing equity through minimum-utility requirements significantly raises guaranteed access levels with only minor reductions in total service. Tightening equity requirements redistributes a portion of the service away from vulnerable groups when resources are scarce, but this effect diminishes as the fleet expands, indicating that equity becomes substantially less costly in resource-rich settings. In order to translate these computational results of vaccination service optimization and planning into actionable insights for decision makers and public health professionals, we used the Levesque Framework as an a posteriori tool to interpret our findings. The framework is helpful in clarifying how barriers in various stages of acccess as described in Levesque

Framework can be addressed by mobile vaccination strategies (including trade-offs between efficiency, resource constraints and equity) for displaced and hard-to-reach populations.

The problem addressed in this study provides a practical, equity-aware approach for large-scale mobile vaccination planning. By integrating fairness considerations with operational constraints, the framework can guide decisions on fleet sizing, service prioritization, and equity targets. It is also compatible with digital health platforms such as Türkiye's national e-NABIZ information system [62], enabling future integration to support personalized communication about mobile vaccination services and improve both utilization and accessibility.

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Appendix A. Operational Mapping of the Levesque Framework to the MFRSP

Table A.1: Operationalization of the Levesque Framework of Access to Healthcare within the proposed MFRSP

Access Dimension	Conceptual Definition	Model-Based Operationalization
Approachability	The extent to which individuals can identify that a service exists, is reachable, and can positively affect their health.	**Identification of Vulnerable Groups:** Explicit classification of 3,292 population groups to detect communities facing information, visibility, or access gaps. **Scale Effects:** Evaluation of how increasing mobile fleet size expands perceived and actual service reach among underserved neighborhoods.
Acceptability	The social and cultural factors that influence individuals' willingness to use services, including trust, beliefs, and perceived appropriateness.	**Uptake Scenarios:** Scenario-based modeling of vaccine hesitancy (low, medium, high show-up rates) to capture behavioral uncertainty. **Priority Weighting:** Higher priority assigned to marginalized populations (e.g., refugees) to mitigate trust deficits and cultural barriers.
Availability and Accommodation	The extent to which services can be reached physically and temporally, considering location, capacity, and organization of delivery.	**Geographic Coverage:** Deployment of MFs to 225 neighborhoods, emphasizing areas without fixed health centers within 1 km. **Temporal Alignment:** Use of Ideal Vaccination Periods (IVPs) and lateness constraints to synchronize service delivery with clinical eligibility and time sensitivity.
Affordability	The capacity of individuals and systems to bear the financial and time-related costs of accessing services, including travel and opportunity costs.	**Operational Cost Control:** Minimization of travel distance and service time to preserve economic feasibility for health authorities. **Equity–Efficiency Trade-off:** Quantification of the system-wide cost associated with enforcing equitable access across groups.
Appropriateness	The degree to which services fit individuals' needs in terms of timing, intensity, and quality of care delivered.	**Minimum Utility Guarantees:** Rawlsian minimum utility constraints ensuring a baseline service level for all groups. **Equity Indices:** Use of Gini and D-index measures to assess and limit systematic disparities in service allocation.

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